
*
* STAAD.Pro
* Version 2006 Bld 1002.US
* Proprietary Program of
* Research Engineers, Intl.
* Date= DEC 26, 2007
* Time= 15:32:58
*
* USER ID: roark
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1. STAAD SPACE BLDG-27-RCC (HEAT EXCHANGER FACILITY)
INPUT FILE: BLDG-27 (26-12-07).STD
2. START JOB INFORMATION
3. ENGINEER DATE 23-OCT-07
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. *****
7. UNIT METER KN
8. JOINT COORDINATES
9. 1 0 -3.9 0; 2 6.5 -2.7 0; 3 12.5 -2 0; 4 19 -2 0; 5 0 -2.7 5.31
10. 6 6.5 -2.7 5.31; 7 12.5 -2 5.31; 8 19 -2 5.31; 9 0 -3.65 11.96
11. 10 6.5 -3.65 11.96; 11 12.5 -2.7 11.96; 12 19 -2 11.96; 13 0 -3.65 17
12. 14 6.5 -3.65 17; 15 12.5 -2.7 17; 16 19 -2 17; 101 0 0 0; 102 6.5 0 0
13. 103 12.5 0 0; 104 19 0 0; 105 0 0 5.31; 106 6.5 0 5.31; 107 12.5 0 5.31
14. 108 19 0 5.31; 109 0 11.96; 110 6.5 0 11.96; 111 12.5 0 11.96
15. 112 19 0 11.96; 113 0 17; 114 6.5 0 17; 115 12.5 0 17; 116 19 0 17
16. 118 0 6.7 0; 119 6.5 6.7 0; 120 12.5 6.7 0; 121 19 6.7 0; 122 0 6.7 5.31
17. 123 6.5 6.7 5.31; 124 12.5 6.7 5.31; 125 19 6.7 5.31; 126 0 6.7 11.96
18. 127 6.5 6.7 11.96; 128 12.5 6.7 11.96; 129 19 6.7 11.96; 130 0 6.7 17
19. 131 6.5 6.7 17; 132 12.5 6.7 17; 133 19 6.7 17; 149 2.975 0 0; 150 0 0 2.425
20. 151 2.975 0 2.425; 152 2.975 0 5.31; 153 0 13.475; 154 6.5 0 13.475
21. 155 12.5 0 14.725; 156 19 0 14.725; 157 16.775 0 17; 158 16.775 0 14.725
22. 159 14.725 0 17; 160 14.725 0 14.725
23. *****
24. MEMBER INCIDENCES
25. 1 101 149; 2 149 102; 3 102 103; 4 103 104; 5 150 151; 6 105 152; 7 152 106
26. 8 106 107; 9 107 108; 10 109 110; 11 110 111; 12 111 112; 13 153 154
27. 14 113 114; 15 114 115; 16 115 159; 17 159 157; 18 157 116; 19 113 153
28. 20 153 109; 21 109 105; 22 105 150; 23 150 101; 24 152 151; 25 151 149
29. 26 114 154; 27 154 110; 28 110 106; 29 106 102; 30 115 155; 31 155 111
30. 32 111 107; 33 107 103; 34 159 160; 35 157 158; 36 116 156; 37 156 112
31. 38 112 108; 39 108 104; 40 155 160; 41 160 158; 42 158 156; 101 118 119
32. 102 119 120; 103 120 121; 104 122 123; 105 123 124; 106 124 125; 107 126 127
33. 108 127 128; 109 128 129; 110 130 131; 111 131 132; 112 132 133; 113 130 126
34. 114 126 122; 115 122 118; 116 131 127; 117 127 123; 118 123 119; 119 132 128
35. 120 128 124; 121 124 120; 122 133 129; 123 129 125; 124 125 121; 1001 1 101
36. 1002 2 102; 1003 3 103; 1004 4 104; 1005 5 105; 1006 6 106; 1007 7 107
37. 1008 8 108; 1009 9 109; 1010 10 110; 1011 11 111; 1012 12 112; 1013 13 113
38. 1014 14 114; 1015 15 115; 1016 16 116; 2001 101 118; 2002 102 119
39. 2003 103 120; 2004 104 121; 2005 105 122; 2008 108 125; 2009 109 126
40. 2010 110 127; 2011 111 128; 2012 112 129; 2013 113 130; 2014 114 131

41. 2015 115 132; 2016 116 133
42. *****
43. DEFINE MATERIAL START
44. ISOTROPIC CONCRETE
45. E 2.17185E+007
46. POISSON 0.17
47. DENSITY 24
48. ALPHA 1E-005
49. DAMP 0.05
50. ISOTROPIC STEEL
51. E 2.05E+008
52. POISSON 0.3
53. DENSITY 78.5
54. ALPHA 1.2E-005
55. DAMP 0.03
56. END DEFINE MATERIAL
57. MEMBER PROPERTY BRITISH
58. *****
59. 1001 TO 1016 PRIS YD 0.55 ZD 0.55
60. 2001 TO 2005 2008 TO 2016 PRIS YD 0.5 ZD 0.5
61. 1 TO 42 PRIS YD 0.6 ZD 0.3
62. 101 TO 124 PRIS YD 0.6 ZD 0.3
63. *****STEEL*****
64. *****
65. CONSTANTS
66. MATERIAL CONCRETE ALL
67. *****
68. SUPPORTS
69. 1 TO 16 PINNED
70. *101 TO 116 118 TO 122 125 TO 133 PINNED
71. *****
72. *MEMBER RELEASE
73. *105 106 START MZ
74. *104 105 END MZ
75. *****
76. UNIT MMS NEWTON
77. MEMBER RELEASE
78. 5 13 24 34 35 40 START MZ
79. 5 13 25 34 35 42 END MZ
80. 18 42 START MX
81. 17 41 END MX
82. UNIT METER KN
83. DEFINE UBC LOAD
84. ZONE 0.15 I 1 RWX 3.5 RWZ 3.5 STYP 4 CT 0.0731 PX 0.514 PZ 0.514 NA 1 NV 1
85. JOINT WEIGHT
86. 101 WEIGHT 357
87. 102 WEIGHT 556
88. 103 WEIGHT 556
89. 104 WEIGHT 399
90. 105 WEIGHT 566
91. 106 WEIGHT 513
92. 107 WEIGHT 506
93. 108 WEIGHT 563
94. 109 WEIGHT 517
95. 110 WEIGHT 708
96. 111 WEIGHT 868

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BLDG-27-RCC (HEAT EXCHANGER FACILITY)

147. 133 FX 0 FY -7.904 FZ 0
 148. 121 FX 0 FY -7.886 FZ 0
 149. 130 FX 0 FY -0.068 FZ -0.786
 150. 118 FX 0 FY -0.068 FZ 0.711
 151. 131 FX 0 FY -0.068 FZ -0.828
 152. 119 FX 0 FY -0.068 FZ 0.774
 153. 132 FX 0 FY -0.068 FZ -0.829
 154. 120 FX 0 FY -0.068 FZ 0.751
 155. 133 FX 0 FY -0.068 FZ -0.789
 156. 121 FX 0 FY -0.068 FZ 0.712
 157. LOAD 2 DL
 158. MEMBER LOAD
 159. *****
 160. *****WALL(T=250 MM,DENSITY=23 KN/M^3,H=6.1 M)=0.25X23X6.1=35.0 KN/M
 161. **PROJECTION OF SLAB (0.62X0.225)=0.62X0.225X25=3.5 KN/M
 162. **FLOOR DL (225 THICK+FIN+M & E)= 7.1 KN/M^2
 163. **ROOF DL (175 THICK+FIN+M & E)= 5.9 KN/M^2
 164. *****
 165. 1 TO 4 14 TO 18 UNI GY -3.5
 166. 19 TO 23 36 TO 39 UNI GY -3.5
 167. *****WALL LOAD *****
 168. 1 TO 4 11 12 14 TO 23 26 27 30 31 36 TO 39 UNI GY -35
 169. FLOOR LOAD
 170. *****G.L.*****
 171. YRANGE -1 1 FLOAD -7.1
 172. YRANGE -1 1 FLOAD 7.1 XRANGE 0 3 ZRANGE 0 2.5
 173. YRANGE -1 1 FLOAD 7.1 XRANGE 0 6.5 ZRANGE 13 17
 174. YRANGE -1 1 FLOAD 7.1 XRANGE 14.5 17 ZRANGE 14.5 17
 175. *****ROOF*****
 176. YRANGE 6 7 FLOAD -5.9
 177. *****
 178. JOINT LOAD
 179. *****DEAD LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 180. 113 FX -0.767 FY -13.107 FZ -1.158
 181. 101 FX -0.415 FY -13.414 FZ 1.177
 182. 114 FX -0.011 FY -18.976 FZ -1.593
 183. 102 FX -0.168 FY -18.395 FZ 1.522
 184. 115 FX 0.011 FY -18.845 FZ -1.579
 185. 103 FX -0.142 FY -18.69 FZ 1.555
 186. 116 FX 0.768 FY -13.069 FZ -1.154
 187. 104 FX 0.623 FY -13.094 FZ 1.158
 188. 126 FX -1.893 FY -33.511 FZ 5.291
 189. 122 FX -1.823 FY -31.788 FZ -5.04
 190. 127 FX -0.004 FY -25.163 FZ -3.342
 191. 123 FX -0.003 FY -23.798 FZ 3.279
 192. 128 FX 0.005 FY -29.124 FZ 2.099
 193. 124 FX 0.005 FY -27.42 FZ -1.965
 194. 129 FX 1.873 FY -34.829 FZ 7.118
 195. 125 FX 1.942 FY -33.009 FZ -6.763
 196. 130 FX 0 FY -24.136 FZ 0
 197. 118 FX 0 FY -24.847 FZ 0
 198. 131 FX 0 FY -30.634 FZ 0
 199. 119 FX 0 FY -31.576 FZ 0

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

97. 112 WEIGHT 661
 98. 113 WEIGHT 298
 99. 114 WEIGHT 543
 100. 115 WEIGHT 625
 101. 116 WEIGHT 380
 102. 118 WEIGHT 124
 103. 119 WEIGHT 358
 104. 120 WEIGHT 360
 105. 121 WEIGHT 122
 106. 122 WEIGHT 298
 107. 125 WEIGHT 301
 108. 126 WEIGHT 228
 109. 127 WEIGHT 542
 110. 128 WEIGHT 546
 111. 129 WEIGHT 230
 112. 130 WEIGHT 125
 113. 131 WEIGHT 147
 114. 132 WEIGHT 146
 115. 133 WEIGHT 125
 116. LOAD 21 EQZ (EARTHQUAKE LOAD)
 117. UBC LOAD X
 118. LOAD 22 EQZ (EARTHQUAKE LOAD)
 119. UBC LOAD Z
 120. *****
 121. LOAD 1 SELFWEIGHT
 122. SELFWEIGHT Y -1
 123. *****SELFWEIGHT TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 124. JOINT LOAD
 125. 113 FX -0.154 FY -5.027 FZ -0.402
 126. 101 FX -0.088 FY -5.149 FZ 0.414
 127. 114 FX -0.006 FY -6.382 FZ -0.509
 128. 102 FX -0.04 FY -6.315 FZ 0.501
 129. 115 FX 0.006 FY -6.382 FZ -0.509
 130. 103 FX -0.029 FY -6.395 FZ 0.511
 131. 116 FX 0.154 FY -5.028 FZ -0.402
 132. 104 FX 0.122 FY -5.078 FZ 0.409
 133. 126 FX 0.159 FY -9.379 FZ 1.464
 134. 122 FX 0.2 FY -8.982 FZ -1.545
 135. 127 FX -0.001 FY -8.777 FZ 1.223
 136. 123 FX -0.001 FY -8.381 FZ -1.296
 137. 128 FX 0.001 FY -8.778 FZ 1.223
 138. 124 FX 0.001 FY -8.378 FZ -1.295
 139. 129 FX -0.163 FY -9.378 FZ 1.468
 140. 125 FX -0.163 FY -8.985 FZ -1.539
 141. 130 FX 0 FY -7.905 FZ 0
 142. 118 FX 0 FY -7.881 FZ 0
 143. 131 FX 0 FY -8.963 FZ 0
 144. 119 FX 0 FY -8.988 FZ 0
 145. 132 FX 0 FY -8.963 FZ 0
 146. 120 FX 0 FY -9.031 FZ 0

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

253. 133 FX 0 FY -12.724 FZ 0
 254. 121 FX 0 FY -13.295 FZ 0
 255. 130 FX 0 FY 0 FZ -3.021
 256. 118 FX 0 FY 0 FZ 3.235
 257. 131 FX 0 FY 0 FZ -5.267
 258. 119 FX 0 FY 0 FZ 5.192
 259. 132 FX 0 FY 0 FZ -4.211
 260. 120 FX 0 FY 0 FZ 4.31
 261. 133 FX 0 FY 0 FZ -2.663
 262. 121 FX 0 FY 0 FZ 2.933
 263. LOAD 4 (NOTIONAL HORIZONTAL LOAD X-DIR)
 264. **PH=1.5% (1.4DL+1.6LL)
 265. JOINT LOAD
 266. 101 FX 8
 267. 102 FX 14
 268. 103 FX 14
 269. 104 FX 10
 270. 105 FX 15
 271. 106 FX 16
 272. 107 FX 16
 273. 108 FX 15
 274. 109 FX 13
 275. 110 FX 19
 276. 111 FX 23
 277. 112 FX 17
 278. 113 FX 7
 279. 114 FX 13
 280. 115 FX 15
 281. 116 FX 9
 282. 118 FX 3
 283. 119 FX 8
 284. 120 FX 8
 285. 121 FX 3
 286. 122 FX 7
 287. 125 FX 7
 288. 126 FX 5
 289. 127 FX 12
 290. 128 FX 12
 291. 129 FX 5
 292. 130 FX 3
 293. 131 FX 3
 294. 132 FX 3
 295. 133 FX 3
 296. LOAD 5 (NOTIONAL HORIZONTAL LOAD (Z-DIR))
 297. **PH=1.5% (1.4DL+1.6LL)
 298. JOINT LOAD
 299. 101 FZ 8
 300. 102 FZ 14
 301. 103 FZ 14
 302. 104 FZ 10
 303. 105 FZ 15
 304. 106 FZ 16
 305. 107 FZ 16
 306. 108 FZ 15
 307. 109 FZ 13
 308. 110 FZ 19

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

200. 132 FX 0 FY -30.253 FZ 0
 201. 120 FX 0 FY -31.275 FZ 0
 202. 133 FX 0 FY -24.005 FZ 0
 203. 121 FX 0 FY -24.765 FZ 0
 204. 130 FX 0 FY 0 FZ -2.101
 205. 118 FX 0 FY 0 FZ 2.355
 206. 131 FX 0 FY 0 FZ -4.569
 207. 119 FX 0 FY 0 FZ 4.491
 208. 132 FX 0 FY 0 FZ -3.134
 209. 120 FX 0 FY 0 FZ 3.252
 210. 133 FX 0 FY 0 FZ -1.633
 211. 121 FX 0 FY 0 FZ 1.942
 212. *****
 213. LOAD 3 LL
 214. MEMBER LOAD
 215. **PROJECTION OF SLAB =0.62X10=6.2 KN/M
 216. **FLOOR LL2 = 10 KN/M^2
 217. **ROOF LL3 = 0.6 KN/M^2
 218. *****
 219. 1 TO 4 TO 14 TO 18 UNI GY -6.2
 220. 19 TO 23 36 TO 39 UNI GY -6.2
 221. FLOOR LOAD
 222. *****G.L.*****
 223. YRANGE -1 1 FLOAD -10
 224. YRANGE -1 1 FLOAD 10 XRANGE 0 3 ZRANGE 0 2.5
 225. YRANGE -1 1 FLOAD 10 XRANGE 0 6.5 ZRANGE 13.4 17
 226. YRANGE -1 1 FLOAD 10 XRANGE 14.5 17 ZRANGE 14.5 17
 227. *****ROOF*****
 228. YRANGE 6 7 FLOAD -0.6
 229. JOINT LOAD
 230. *****LIVE LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 231. 113 FX -0.029 FY 1.422 FZ 0.196
 232. 101 FX -0.006 FY 1.623 FZ -0.206
 233. 114 FX 0.008 FY 1.47 FZ 0.206
 234. 102 FX 0.007 FY 1.586 FZ -0.22
 235. 115 FX -0.008 FY 1.564 FZ 0.216
 236. 103 FX -0.004 FY 1.651 FZ -0.228
 237. 116 FX 0.03 FY 1.455 FZ 0.2
 238. 104 FX 0.028 FY 1.497 FZ -0.206
 239. 126 FX -1.415 FY -25.094 FZ 3.298
 240. 122 FX -1.43 FY -23.837 FZ -3.079
 241. 127 FX 0 FY -18.917 FZ -3.336
 242. 123 FX -0.001 FY -17.912 FZ 3.288
 243. 128 FX 0.001 FY -21.85 FZ 0.701
 244. 124 FX 0 FY -20.605 FZ -0.583
 245. 129 FX 1.418 FY -26.072 FZ 4.641
 246. 125 FX 1.402 FY -24.73 FZ -4.379
 247. 130 FX 0 FY -12.817 FZ 0
 248. 118 FX 0 FY -13.507 FZ 0
 249. 131 FX 0 FY -15.389 FZ 0
 250. 119 FX 0 FY -16.127 FZ 0
 251. 132 FX 0 FY -15.104 FZ 0
 252. 120 FX 0 FY -15.768 FZ 0

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

365. LOAD 12 WL2 (0 DEG , INTERNAL PRESSURE +0.2)
 366. JOINT LOAD
 367. 113 FX 0.078 FY 29.442 FZ 10.688
 368. 101 FX -0.037 FY -33.735 FZ 9.067
 369. 114 FX -0.003 FY 38.854 FZ 14.015
 370. 102 FX -0.034 FY -34.865 FZ 11.868
 371. 115 FX 0.003 FY 38.696 FZ 13.999
 372. 103 FX -0.011 FY -35.049 FZ 11.89
 373. 116 FX -0.078 FY 29.399 FZ 10.684
 374. 104 FX -0.026 FY -27.587 FZ 9.198
 375. 126 FX 1.16 FY 24.46 FZ -8.189
 376. 122 FX 2.951 FY 44.451 FZ 1.92
 377. 127 FX 0 FY 16.787 FZ 0.591
 378. 123 FX 0.005 FY 34.742 FZ -7.583
 379. 128 FX -0.001 FY 20.939 FZ -5.067
 380. 124 FX -0.004 FY 38.558 FZ -2.095
 381. 129 FX -1.14 FY 25.852 FZ -10.112
 382. 125 FX -2.863 FY 45.677 FZ 3.742
 383. 130 FX 0 FY -20.623 FZ 0
 384. 118 FX 0 FY 57.188 FZ 0
 385. 131 FX 0 FY -27.716 FZ 0
 386. 119 FX 0 FY 63.741 FZ 0
 387. 132 FX 0 FY -28.099 FZ 0
 388. 120 FX 0 FY 63.15 FZ 0
 389. 133 FX 0 FY -20.758 FZ 0
 390. 121 FX 0 FY 51.057 FZ 0
 391. 130 FX 0 FY 0 FZ -14.168
 392. 118 FX 0 FY 0 FZ -20.523
 393. 131 FX 0 FY 0 FZ -15.506
 394. 119 FX 0 FY 0 FZ -27.999
 395. 132 FX 0 FY 0 FZ -17.06
 396. 120 FX 0 FY 0 FZ -26.871
 397. 133 FX 0 FY 0 FZ -14.67
 398. 121 FX 0 FY 0 FZ -19.988
 399. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 400. LOAD 13 WL3 (90 DEG , INTERNAL SUCTION -0.3, WM&LW +0.2)
 401. JOINT LOAD
 402. 113 FX 0.011 FY 23.471 FZ 7.955
 403. 101 FX 0.022 FY 28.783 FZ -7.846
 404. 114 FX 0.01 FY 30.07 FZ 10.321
 405. 102 FX 0.021 FY 30.088 FZ -10.323
 406. 115 FX -0.01 FY 30.165 FZ 10.331
 407. 103 FX 0.006 FY 30.231 FZ -10.339
 408. 116 FX -0.011 FY 23.492 FZ 7.957
 409. 104 FX -0.001 FY 23.514 FZ -7.961
 410. 126 FX -1.459 FY -25.855 FZ 3.322
 411. 122 FX -1.487 FY -24.628 FZ -3.143
 412. 127 FX -0.001 FY -19.532 FZ -3.477
 413. 123 FX -0.002 FY -18.528 FZ 3.402
 414. 128 FX 0.001 FY -22.538 FZ 0.662
 415. 124 FX 0 FY -21.291 FZ -0.563
 416. 129 FX 1.466 FY -26.861 FZ 4.735
 417. 125 FX 1.434 FY -25.509 FZ -4.483
 418. 130 FX 0 FY -34.632 FZ 0
 419. 118 FX 0 FY -40.347 FZ 0
 420. 131 FX 0 FY -43.839 FZ 0

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

309. 111 FZ 23
 310. 112 FZ 17
 311. 113 FZ 7
 312. 114 FZ 13
 313. 115 FZ 15
 314. 116 FZ 8
 315. 118 FZ 3
 316. 119 FZ 8
 317. 120 FZ 8
 318. 121 FZ 3
 319. 122 FZ 7
 320. 125 FZ 7
 321. 126 FZ 5
 322. 127 FZ 12
 323. 128 FZ 12
 324. 129 FZ 5
 325. 130 FZ 3
 326. 131 FZ 3
 327. 132 FZ 3
 328. 133 FZ 3
 329. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 330. LOAD 11 WL1 (0 DEG , INTERNAL SUCTION -0.3)
 331. JOINT LOAD
 332. 113 FX -0.013 FY 9.734 FZ 3.238
 333. 101 FX -0.069 FY -59.253 FZ 16.499
 334. 114 FX 0.009 FY 12.566 FZ 4.207
 335. 102 FX -0.042 FY -61.947 FZ 21.772
 336. 115 FX -0.009 FY 12.563 FZ 4.206
 337. 103 FX -0.025 FY -62.121 FZ 21.792
 338. 116 FX 0.013 FY 9.764 FZ 3.242
 339. 104 FX 0.03 FY -48.056 FZ 16.743
 340. 126 FX -1.467 FY -18.773 FZ -5.269
 341. 122 FX 1.499 FY 17.639 FZ -5.131
 342. 127 FX -0.001 FY -16.637 FZ -6.283
 343. 123 FX 0.004 FY 15.65 FZ -5.823
 344. 128 FX 0.001 FY -16.696 FZ -6.19
 345. 124 FX -0.002 FY 15.605 FZ -5.88
 346. 129 FX 1.467 FY -18.784 FZ -5.322
 347. 125 FX -1.396 FY 17.539 FZ -5.141
 348. 130 FX 0 FY -21.959 FZ 0
 349. 118 FX 0 FY 70.832 FZ 0
 350. 131 FX 0 FY -26.802 FZ 0
 351. 119 FX 0 FY 76.204 FZ 0
 352. 132 FX 0 FY -26.793 FZ 0
 353. 120 FX 0 FY 76.069 FZ 0
 354. 133 FX 0 FY -21.981 FZ 0
 355. 121 FX 0 FY 59.878 FZ 0
 356. 130 FX 0 FY 0 FZ -10.23
 357. 118 FX 0 FY 0 FZ -28.616
 358. 131 FX 0 FY 0 FZ -12.193
 359. 119 FX 0 FY 0 FZ -36.24
 360. 132 FX 0 FY 0 FZ -12.185
 361. 120 FX 0 FY 0 FZ -36.407
 362. 133 FX 0 FY 0 FZ -10.159
 363. 121 FX 0 FY 0 FZ -28.495
 364. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

421. 119 FX 0 FY -44.553 FZ 0
 422. 132 FX 0 FY -43.546 FZ 0
 423. 120 FX 0 FY -44.147 FZ 0
 424. 133 FX 0 FY -34.526 FZ 0
 425. 121 FX 0 FY -35.07 FZ 0
 426. 130 FX 0 FY 0 FZ -13.417
 427. 118 FX 0 FY 0 FZ 13.712
 428. 131 FX 0 FY 0 FZ -18.921
 429. 119 FX 0 FY 0 FZ 18.819
 430. 132 FX 0 FY 0 FZ -17.84
 431. 120 FX 0 FY 0 FZ 17.924
 432. 133 FX 0 FY 0 FZ -13.08
 433. 121 FX 0 FY 0 FZ 13.352
 434. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 435. LOAD 14 WL4 (90 DEG , INTERNAL PRESSURE +0.2, WW&LW -0.7)
 436. JOINT LOAD
 437. 113 FX 0.126 FY 41.694 FZ 15.203
 438. 101 FX 0.064 FY 51.59 FZ -14.951
 439. 114 FX -0.009 FY 55.069 FZ 19.946
 440. 102 FX 0.016 FY 54.681 FZ -19.9
 441. 115 FX 0.009 FY 54.877 FZ 19.926
 442. 103 FX 0.027 FY 54.701 FZ -19.9
 443. 116 FX -0.127 FY 41.588 FZ 15.191
 444. 104 FX -0.105 FY 41.491 FZ -15.178
 445. 126 FX 2.647 FY 45.953 FZ -7.033
 446. 122 FX 2.663 FY 43.569 FZ 6.524
 447. 127 FX 0.002 FY 34.346 FZ 5.145
 448. 123 FX 0.002 FY 32.503 FZ -5.15
 449. 128 FX -0.003 FY 39.758 FZ -2.264
 450. 124 FX -0.002 FY 37.469 FZ 1.958
 451. 129 FX -2.653 FY 47.756 FZ -9.409
 452. 125 FX -2.657 FY 45.296 FZ 8.876
 453. 130 FX 0 FY -20.826 FZ 0
 454. 118 FX 0 FY -29.653 FZ 0
 455. 131 FX 0 FY -29.621 FZ 0
 456. 119 FX 0 FY -28.271 FZ 0
 457. 132 FX 0 FY -30.132 FZ 0
 458. 120 FX 0 FY -28.832 FZ 0
 459. 133 FX 0 FY -20.962 FZ 0
 460. 121 FX 0 FY -19.888 FZ 0
 461. 130 FX 0 FY 0 FZ -15.307
 462. 118 FX 0 FY 0 FZ 15.049
 463. 131 FX 0 FY 0 FZ -16.814
 464. 119 FX 0 FY 0 FZ 16.843
 465. 132 FX 0 FY 0 FZ -18.805
 466. 120 FX 0 FY 0 FZ 18.537
 467. 133 FX 0 FY 0 FZ -16.068
 468. 121 FX 0 FY 0 FZ 15.549
 469. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 470. LOAD 15 WL5 (0 DEG , INTERNAL SUCTION -0.3, LEE WARD -0.7, REVERSE)
 471. JOINT LOAD
 472. 113 FX -0.067 FY -48.036 FZ -16.739
 473. 101 FX 0.005 FY 12.045 FZ -3.204
 474. 114 FX -0.009 FY -62.129 FZ -21.795
 475. 102 FX 0.013 FY 12.55 FZ -4.204
 476. 115 FX 0.009 FY -62.129 FZ -21.795

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

477. 103 FX 0 FY 12.609 FZ -4.213
 478. 116 FX 0.067 FY -48.039 FZ -16.74
 479. 104 FX 0.016 FY 9.823 FZ -3.25
 480. 126 FX 1.451 FY 18.022 FZ 5.818
 481. 122 FX -1.484 FY -18.256 FZ 4.686
 482. 127 FX 0.003 FY 16.18 FZ 6.615
 483. 123 FX -0.002 FY -16.051 FZ 5.502
 484. 128 FX -0.003 FY 16.122 FZ 6.699
 485. 124 FX 0 FY -16.108 FZ 5.431
 486. 129 FX -1.445 FY 18 FZ 5.857
 487. 125 FX 1.446 FY -18.256 FZ 4.65
 488. 130 FX 0 FY 59.36 FZ 0
 489. 118 FX 0 FY -24.708 FZ 0
 490. 131 FX 0 FY 75.628 FZ 0
 491. 119 FX 0 FY -27.439 FZ 0
 492. 132 FX 0 FY 75.635 FZ 0
 493. 120 FX 0 FY -27.392 FZ 0
 494. 133 FX 0 FY 59.365 FZ 0
 495. 121 FX 0 FY -22.553 FZ 0
 496. 130 FX 0 FY 0 FZ 28.46
 497. 118 FX 0 FY 0 FZ 10.268
 498. 131 FX 0 FY 0 FZ 36.289
 499. 119 FX 0 FY 0 FZ 12.243
 500. 132 FX 0 FY 0 FZ 36.307
 501. 120 FX 0 FY 0 FZ 12.239
 502. 133 FX 0 FY 0 FZ 28.459
 503. 121 FX 0 FY 0 FZ 10.238
 504. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 505. LOAD 16 WL6 (0 DEG , INTERNAL PRESSURE +0.2, LEE WARD -0.2, REVERSE)
 506. JOINT LOAD
 507. 113 FX -0.001 FY -27.449 FZ -9.18
 508. 101 FX 0.045 FY 36.52 FZ -10.52
 509. 114 FX -0.014 FY -34.803 FZ -11.861
 510. 102 FX 0.015 FY 38.577 FZ -13.982
 511. 115 FX 0.014 FY -34.948 FZ -11.876
 512. 103 FX 0.018 FY 38.593 FZ -13.983
 513. 116 FX 0 FY -27.528 FZ -9.188
 514. 104 FX -0.061 FY 29.363 FZ -10.678
 515. 126 FX 2.897 FY 46.462 FZ -1.877
 516. 122 FX 1.155 FY 22.943 FZ 7.512
 517. 127 FX 0.004 FY 36.472 FZ 8.075
 518. 123 FX 0 FY 15.713 FZ -1.041
 519. 128 FX -0.005 FY 40.622 FZ 2.389
 520. 124 FX -0.001 FY 19.521 FZ 4.405
 521. 129 FX -2.905 FY 47.846 FZ -3.705
 522. 125 FX -1.163 FY 24.265 FZ 9.314
 523. 130 FX 0 FY 50.056 FZ 0
 524. 118 FX 0 FY -27.166 FZ 0
 525. 131 FX 0 FY 62.373 FZ 0
 526. 119 FX 0 FY -27.052 FZ 0
 527. 132 FX 0 FY 61.98 FZ 0
 528. 120 FX 0 FY -27.468 FZ 0
 529. 133 FX 0 FY 49.949 FZ 0
 530. 121 FX 0 FY -20.279 FZ 0
 531. 130 FX 0 FY 0 FZ 20.165
 532. 118 FX 0 FY 0 FZ 13.929

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

533. 131 FX 0 FY 0 FZ 28.126
 534. 119 FX 0 FY 0 FZ 15.579
 535. 132 FX 0 FY 0 FZ 26.603
 536. 120 FX 0 FY 0 FZ 16.905
 537. 133 FX 0 FY 0 FZ 19.586
 538. 121 FX 0 FY 0 FZ 14.314
 539. *****
 540. ** SERVICEABILITY*
 541. *****
 542. LOAD COMB 101 1.0DL+1.0LL
 543. 1 1.0 2 1.0 3 1.0
 544. LOAD COMB 102 1.0DL+0.5LL
 545. 1 1.0 2 1.0 3 0.5
 546. LOAD COMB 103 0.8DL+0.8LL+0.8WL1
 547. 1 0.8 2 0.8 3 0.8 11 0.8
 548. LOAD COMB 104 1.0DL+1.0WL1
 549. 1 1.0 2 1.0 11 1.0
 550. LOAD COMB 105 0.8DL+0.8LL+0.8WL2
 551. 1 0.8 2 0.8 3 0.8 12 0.8
 552. LOAD COMB 106 1.0DL+1.0WL2
 553. 1 1.0 2 1.0 12 1.0
 554. LOAD COMB 107 0.8DL+0.8LL+0.8WL3
 555. 1 0.8 2 0.8 3 0.8 13 0.8
 556. LOAD COMB 108 1.0DL+1.0WL3
 557. 1 1.0 2 1.0 13 1.0
 558. LOAD COMB 109 0.8DL+0.8LL+0.8WL4
 559. 1 0.8 2 0.8 3 0.8 14 0.8
 560. LOAD COMB 110 1.0DL+1.0WL4
 561. 1 1.0 2 1.0 14 1.0
 562. LOAD COMB 111 0.8DL+0.8LL+0.8WL5
 563. 1 0.8 2 0.8 3 0.8 15 0.8
 564. LOAD COMB 112 1.0DL+1.0WL4
 565. 1 1.0 2 1.0 15 1.0
 566. LOAD COMB 113 0.8DL+0.8LL+0.8WL6
 567. 1 0.8 2 0.8 3 0.8 16 0.8
 568. LOAD COMB 114 1.0DL+1.0WL6
 569. 1 1.0 2 1.0 16 1.0
 570. *****EQX & EQZ *****
 571. **DL+EQ/1.4 = 1.0DL+0.72EQX+0.22 EQZ, EQ=EQX+0.3EQZ
 572. **0.9 DL+EQ/1.4 =0.9 DL+0.72 EQX+0.22EQZ
 573. **DL 0.75(LL+EQ/1.4) =1.0 DL+0.75 LL+0.54 EQX+0.17EQZ
 574. *****
 575. LOAD COMB 115 DL+0.72EQX+0.22EQZ
 576. 1 1.0 2 1.0 21 0.72 22 0.22
 577. LOAD COMB 116 DL+0.72EQX-0.22EQZ
 578. 1 1.0 2 1.0 21 0.72 22 -0.22
 579. LOAD COMB 117 DL-0.72EQX+0.22EQZ
 580. 1 1.0 2 1.0 21 -0.72 22 0.22
 581. LOAD COMB 118 DL-0.72EQX+0.22EQZ
 582. 1 1.0 2 1.0 21 -0.72 22 -0.22
 583. LOAD COMB 119 DL+0.72EQX+0.22EQX
 584. 1 1.0 2 1.0 22 0.72 21 0.22
 585. LOAD COMB 120 DL+0.72EQZ-0.22EQX
 586. 1 1.0 2 1.0 22 0.72 21 -0.22
 587. LOAD COMB 121 DL-0.72EQZ+0.22EQX
 588. 1 1.0 2 1.0 22 -0.72 21 0.22

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

589. LOAD COMB 122 DL-0.72EQZ-0.22EQX
 590. 1 1.0 2 1.0 22 -0.72 21 -0.22
 591. LOAD COMB 123 0.9DL+0.72EQX+0.22EQZ
 592. 1 0.9 2 0.9 21 0.72 22 0.22
 593. LOAD COMB 124 0.9DL+0.72EQX-0.22EQZ
 594. 1 0.9 2 0.9 21 0.72 22 -0.22
 595. LOAD COMB 125 0.9DL-0.72EQX+0.22EQZ
 596. 1 0.9 2 0.9 21 -0.72 22 0.22
 597. LOAD COMB 126 0.9DL-0.72EQX+0.22EQZ
 598. 1 0.9 2 0.9 21 -0.72 22 -0.22
 599. LOAD COMB 127 0.9DL+0.72EQZ+0.22EQX
 600. 1 0.9 2 0.9 22 0.72 21 0.22
 601. LOAD COMB 128 0.9DL+0.72EQZ-0.22EQX
 602. 1 0.9 2 0.9 22 0.72 21 -0.22
 603. LOAD COMB 129 0.9DL-0.72EQZ+0.22EQX
 604. 1 0.9 2 0.9 22 -0.72 21 0.22
 605. LOAD COMB 130 0.9DL-0.72EQZ-0.22EQX
 606. 1 0.9 2 0.9 22 -0.72 21 -0.22
 607. LOAD COMB 131 1.0DL+0.75LL+0.54EQX+0.17 EQZ
 608. 1 1.0 2 1.0 3 0.75 21 0.54 22 0.17
 609. LOAD COMB 132 1.0DL+0.75LL+0.54EQX-0.17 EQZ
 610. 1 1.0 2 1.0 3 0.75 21 0.54 22 -0.17
 611. LOAD COMB 133 1.0DL+0.75LL-0.54EQX+0.17 EQZ
 612. 1 1.0 2 1.0 3 0.75 21 -0.54 22 0.17
 613. LOAD COMB 134 1.0DL+0.75LL-0.54EQX-0.17 EQZ
 614. 1 1.0 2 1.0 3 0.75 21 -0.54 22 -0.17
 615. LOAD COMB 135 1.0DL+0.75LL+0.54EQZ+0.17 EQX
 616. 1 1.0 2 1.0 3 0.75 22 0.54 21 0.17
 617. LOAD COMB 136 1.0DL+0.75LL+0.54EQZ-0.17 EQX
 618. 1 1.0 2 1.0 3 0.75 22 0.54 21 -0.17
 619. LOAD COMB 137 1.0DL+0.75LL-0.54EQZ+0.17 EQX
 620. 1 1.0 2 1.0 3 0.75 22 -0.54 21 0.17
 621. LOAD COMB 138 1.0DL+0.75LL-0.54EQZ-0.17 EQX
 622. 1 1.0 2 1.0 3 0.75 22 -0.54 21 -0.17
 623. *****
 624. ** CONCRETE DESIGN
 625. *****
 626. LOAD COMB 200 1.4DL+1.6LL
 627. 1 1.4 2 1.4 3 1.6
 628. *****WL1*****
 629. LOAD COMB 201 1.2DL+1.2LL+1.2WL1
 630. 1 1.2 2 1.2 3 1.2 11 1.2
 631. LOAD COMB 202 1.4DL+1.4WL1
 632. 1 1.4 2 1.4 11 1.4
 633. LOAD COMB 203 1DL+1.4WL1
 634. 1 1.0 2 1.0 11 1.4
 635. *****WL2*****
 636. LOAD COMB 204 1.2DL+1.2LL+1.2WL2
 637. 1 1.2 2 1.2 3 1.2 12 1.2
 638. LOAD COMB 205 1.4DL+1.4WL2
 639. 1 1.4 2 1.4 12 1.4
 640. LOAD COMB 206 1DL+1.4WL2
 641. 1 1.0 2 1.0 12 1.4
 642. *****WL3*****
 643. LOAD COMB 207 1.2DL+1.2LL+1.2WL3
 644. 1 1.2 2 1.2 3 1.2 13 1.2

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

645. LOAD COMB 208 1.4DL+1.4WL3
 646. 1 1.4 2 1.4 13 1.4
 647. LOAD COMB 209 1DL+1.4WL3
 648. 1 1.0 2 1.0 13 1.4
 649. *****WL4*****
 650. LOAD COMB 210 1.2DL+1.2LL+1.2WL4
 651. 1 1.2 2 1.2 3 1.2 14 1.2
 652. LOAD COMB 211 1.4DL+1.4WL4
 653. 1 1.4 2 1.4 14 1.4
 654. LOAD COMB 212 1DL+1.4WL4
 655. 1 1.0 2 1.0 14 1.4
 656. *****WL5*****
 657. LOAD COMB 213 1.2DL+1.2LL+1.2WL5
 658. 1 1.2 2 1.2 3 1.2 15 1.2
 659. LOAD COMB 214 1.4DL+1.4WL5
 660. 1 1.4 2 1.4 15 1.4
 661. LOAD COMB 215 1DL+1.4WL5
 662. 1 1.0 2 1.0 15 1.4
 663. *****WL6*****
 664. LOAD COMB 216 1.2DL+1.2LL+1.2WL6
 665. 1 1.2 2 1.2 3 1.2 16 1.2
 666. LOAD COMB 217 1.4DL+1.4WL6
 667. 1 1.4 2 1.4 16 1.4
 668. LOAD COMB 218 1DL+1.4WL6
 669. 1 1.0 2 1.0 16 1.4
 670. *****EQX & EQZ *****
 671. LOAD COMB 219 1.32DL+1.1LL+1.1EQX+0.33 EQZ
 672. 1 1.32 2 1.32 3 1.1 21 1.1 22 0.33
 673. LOAD COMB 220 1.32DL+1.1LL+1.1EQX-0.33 EQZ
 674. 1 1.32 2 1.32 3 1.1 21 1.1 22 -0.33
 675. LOAD COMB 221 1.32DL+1.1LL-1.1EQX+0.33 EQZ
 676. 1 1.32 2 1.32 3 1.1 21 -1.1 22 0.33
 677. LOAD COMB 222 1.32DL+1.1LL-1.1EQX-0.33 EQZ
 678. 1 1.32 2 1.32 3 1.1 21 -1.1 22 -0.33
 679. LOAD COMB 223 1.32DL+1.1LL+1.1EQZ+0.33 EQX
 680. 1 1.32 2 1.32 3 1.1 22 1.1 21 0.33
 681. LOAD COMB 224 1.32DL+1.1LL+1.1EQZ-0.33 EQX
 682. 1 1.32 2 1.32 3 1.1 22 1.1 21 -0.33
 683. LOAD COMB 225 1.32DL+1.1LL-1.1EQZ+0.33 EQX
 684. 1 1.32 2 1.32 3 1.1 22 -1.1 21 0.33
 685. LOAD COMB 226 1.32DL+1.1LL-1.1EQZ-0.33 EQX
 686. 1 1.32 2 1.32 3 1.1 22 -1.1 21 -0.33
 687. LOAD COMB 227 0.99DL+1.1EQX+0.33EQZ
 688. 1 0.99 2 0.99 21 1.1 22 0.33
 689. LOAD COMB 228 0.99DL+1.1EQX-0.33EQZ
 690. 1 0.99 2 0.99 21 1.1 22 -0.33
 691. LOAD COMB 229 0.99DL-1.1EQX+0.33EQZ
 692. 1 0.99 2 0.99 21 -1.1 22 0.33
 693. LOAD COMB 230 0.99DL-1.1EQX-0.33EQZ
 694. 1 0.99 2 0.99 21 -1.1 22 -0.33
 695. LOAD COMB 231 0.99DL+1.1EQZ+0.33EQX
 696. 1 0.99 2 0.99 22 1.1 21 0.33
 697. LOAD COMB 232 0.99DL+1.1EQZ-0.33EQX
 698. 1 0.99 2 0.99 22 1.1 21 -0.33
 699. LOAD COMB 233 0.99DL-1.1EQZ+0.33EQX
 700. 1 0.99 2 0.99 22 -1.1 21 0.33

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

701. LOAD COMB 234 0.99DL-1.1EQZ-0.33EQX
 702. 1 0.99 2 0.99 22 -1.1 21 -0.33
 703. LOAD COMB 235 1.0DL+1.0LL+1.0PHX
 704. 1 1.0 2 1.0 3 1.0 4 1.0
 705. LOAD COMB 236 1.0DL+1.0LL+1.0PHZ
 706. 1 1.0 2 1.0 3 1.0 5 1.0
 707. *****DEFLECTION*****
 708. LOAD COMB 301 SELF+DL
 709. 1 1.0 2 1.0
 710. LOAD COMB 302 LL
 711. 3 1.0
 712. LOAD COMB 303 0.8 (LL+WL1)
 713. 3 0.8 11 0.8
 714. LOAD COMB 304 0.8 (LL+WL2)
 715. 3 0.8 12 0.8
 716. LOAD COMB 305 0.8 (LL+WL3)
 717. 3 0.8 13 0.8
 718. LOAD COMB 306 0.8 (LL+WL4)
 719. 3 0.8 14 0.8
 720. LOAD COMB 307 0.8 (LL+WL5)
 721. 3 0.8 15 0.8
 722. LOAD COMB 308 0.8 (LL+WL6)
 723. 3 0.8 16 0.8
 724. LOAD COMB 309 0.8 (LL+EQX/1.4+0.3EQZ/1.4)
 725. 3 0.8 21 0.57 22 0.17
 726. LOAD COMB 310 0.8 (LL+EQX/1.4-0.3EQZ/1.4)
 727. 3 0.8 21 0.57 22 -0.17
 728. LOAD COMB 311 0.8 (LL-EQX/1.4+0.3EQZ/1.4)
 729. 3 0.8 21 -0.57 22 0.17
 730. LOAD COMB 312 0.8 (LL-EQX/1.4-0.3EQZ/1.4)
 731. 3 0.8 21 -0.57 22 -0.17
 732. LOAD COMB 313 0.8 (LL+EQZ/1.4+0.3EQX/1.4)
 733. 3 0.8 22 0.57 21 0.17
 734. LOAD COMB 314 0.8 (LL+EQZ/1.4-0.3EQX/1.4)
 735. 3 0.8 22 0.57 21 -0.17
 736. LOAD COMB 315 0.8 (LL-EQZ/1.4+0.3EQX/1.4)
 737. 3 0.8 21 -0.57 22 0.17
 738. LOAD COMB 316 0.8 (LL-EQZ/1.4-0.3EQX/1.4)
 739. 3 0.8 21 -0.57 22 -0.17
 740. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 60/ 96/ 16
 ORIGINAL/FINAL BAND-WIDTH= 33/ 12/ 66 DOF
 TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 312
 SIZE OF STIFFNESS MATRIX = 21 DOUBLE KILO-WORDS
 REQD/AVAIL. DISK SPACE = 12.5/ 22160.8 MB

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS & CHANGE.

ZERO STIFFNESS IN DIRECTION 4 AT JOINT 157 EQN.NO. 4
LOADS APPLIED OR DISTRIBUTED HERE FROM ELEMENTS WILL BE IGNORED.
THIS MAY BE DUE TO ALL MEMBERS AT THIS JOINT BEING RELEASED OR
EFFECTIVELY RELEASED IN THIS DIRECTION.

ZERO STIFFNESS IN DIRECTION 4 AT JOINT 158 EQN.NO. 22

*

* X DIRECTION : Ta = 0.429 Tb = 0.811 Tuser = 0.514 *

* T = 0.514, LOAD FACTOR = 1.000 *

* UBC TYPE = 97 *

* UBC FACTOR V = 0.1571 x 12268.00 = 1927.83 KN *

*

*

* Z DIRECTION : Ta = 0.429 Tb = 0.796 Tuser = 0.514 *

* T = 0.514, LOAD FACTOR = 1.000 *

* UBC TYPE = 97 *

* UBC FACTOR V = 0.1571 x 12268.00 = 1927.83 KN *

*

741. LOAD LIST 101 TO 138

742. PRINT SUPPORT REACTION

□ SUPPORT REACTION

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1	101	34.73	536.15	11.57	0.00	0.00	0.00
	102	29.73	493.99	10.80	0.00	0.00	0.00
	103	27.67	446.99	11.53	0.00	0.00	0.00
	104	24.58	474.43	12.86	0.00	0.00	0.00
	105	27.73	432.60	10.87	0.00	0.00	0.00
	106	24.65	456.45	12.04	0.00	0.00	0.00
	107	27.76	437.43	9.29	0.00	0.00	0.00
	108	24.69	462.48	10.06	0.00	0.00	0.00
	109	27.82	411.91	9.54	0.00	0.00	0.00
	110	24.77	430.58	10.38	0.00	0.00	0.00
	111	27.90	412.06	6.82	0.00	0.00	0.00
	112	24.86	430.76	6.98	0.00	0.00	0.00
	113	27.90	400.41	7.83	0.00	0.00	0.00
	114	24.87	416.20	8.23	0.00	0.00	0.00
	115	20.55	299.01	-0.90	0.00	0.00	0.00
	116	20.06	395.15	23.49	0.00	0.00	0.00
	117	29.39	508.53	-3.44	0.00	0.00	0.00
	118	28.90	604.68	20.95	0.00	0.00	0.00
	119	24.17	262.51	-29.50	0.00	0.00	0.00
	120	26.87	326.53	-30.28	0.00	0.00	0.00
	121	22.57	577.15	50.32	0.00	0.00	0.00
	122	25.27	641.17	49.54	0.00	0.00	0.00
	123	18.07	253.82	-1.90	0.00	0.00	0.00
	124	17.59	349.96	22.49	0.00	0.00	0.00
	125	26.91	463.35	-4.45	0.00	0.00	0.00
	126	26.42	559.49	19.94	0.00	0.00	0.00
	127	21.70	217.32	-30.50	0.00	0.00	0.00
	128	24.40	281.34	-31.28	0.00	0.00	0.00
	129	20.10	531.97	49.32	0.00	0.00	0.00
	130	22.80	595.99	48.54	0.00	0.00	0.00
	131	29.10	399.35	2.71	0.00	0.00	0.00
	132	28.72	473.64	21.56	0.00	0.00	0.00
	133	35.73	556.50	0.81	0.00	0.00	0.00
	134	35.35	630.79	19.65	0.00	0.00	0.00
	135	31.78	372.34	-18.45	0.00	0.00	0.00
	136	33.87	421.81	-19.05	0.00	0.00	0.00
	137	30.58	608.33	41.42	0.00	0.00	0.00
	138	32.67	657.80	40.82	0.00	0.00	0.00
2	101	-5.30	1031.04	-5.33	0.00	0.00	0.00
	102	-5.09	916.99	-8.41	0.00	0.00	0.00
	103	-4.66	826.44	-1.97	0.00	0.00	0.00
	104	-5.39	804.93	-8.63	0.00	0.00	0.00
	105	-4.55	803.30	-2.11	0.00	0.00	0.00
	106	-5.25	776.01	-8.81	0.00	0.00	0.00
	107	-4.25	843.58	-4.84	0.00	0.00	0.00
	108	-4.88	826.36	-12.22	0.00	0.00	0.00
	109	-4.23	790.96	-2.25	0.00	0.00	0.00
	110	-4.86	760.58	-8.99	0.00	0.00	0.00

□

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
111	-3.82	823.99	-7.04	0.00	0.00	0.00	0.00
112	-4.35	801.88	-14.96	0.00	0.00	0.00	0.00
113	-3.91	794.40	-4.05	0.00	0.00	0.00	0.00
114	-4.46	764.89	-11.23	0.00	0.00	0.00	0.00
115	-78.07	768.71	-39.01	0.00	0.00	0.00	0.00
116	-79.96	852.47	11.12	0.00	0.00	0.00	0.00
117	70.22	753.39	-34.12	0.00	0.00	0.00	0.00
118	68.33	837.15	16.01	0.00	0.00	0.00	0.00
119	-24.44	668.21	-94.27	0.00	0.00	0.00	0.00
120	20.87	663.53	-92.78	0.00	0.00	0.00	0.00
121	-30.61	942.33	69.78	0.00	0.00	0.00	0.00
122	14.70	937.65	71.28	0.00	0.00	0.00	0.00
123	-77.58	688.42	-37.86	0.00	0.00	0.00	0.00
124	-79.47	772.18	12.27	0.00	0.00	0.00	0.00
125	70.70	673.09	-32.97	0.00	0.00	0.00	0.00
126	68.82	756.85	17.16	0.00	0.00	0.00	0.00
127	-23.95	587.92	-93.12	0.00	0.00	0.00	0.00
128	21.36	583.23	-91.63	0.00	0.00	0.00	0.00
129	-30.12	862.04	70.93	0.00	0.00	0.00	0.00
130	15.19	857.36	72.43	0.00	0.00	0.00	0.00
131	-60.07	947.40	-28.07	0.00	0.00	0.00	0.00
132	-61.53	1012.12	10.66	0.00	0.00	0.00	0.00
133	51.14	935.91	-24.40	0.00	0.00	0.00	0.00
134	49.69	1000.63	14.33	0.00	0.00	0.00	0.00
135	-20.39	873.03	-68.97	0.00	0.00	0.00	0.00
136	14.62	869.41	-67.81	0.00	0.00	0.00	0.00
137	-25.01	1078.62	54.07	0.00	0.00	0.00	0.00
138	10.00	1075.00	55.23	0.00	0.00	0.00	0.00
101	40.99	1040.89	-2.50	0.00	0.00	0.00	0.00
102	34.04	920.45	-8.45	0.00	0.00	0.00	0.00
103	31.77	834.26	-1.63	0.00	0.00	0.00	0.00
104	25.82	801.93	-13.92	0.00	0.00	0.00	0.00
105	31.95	810.83	-0.75	0.00	0.00	0.00	0.00
106	26.04	772.65	-12.83	0.00	0.00	0.00	0.00
107	32.86	851.66	-2.80	0.00	0.00	0.00	0.00
108	27.18	823.69	-15.39	0.00	0.00	0.00	0.00
109	32.70	798.22	1.26	0.00	0.00	0.00	0.00
110	26.98	756.89	-10.32	0.00	0.00	0.00	0.00
111	33.81	832.06	-3.26	0.00	0.00	0.00	0.00
112	28.37	799.19	-15.96	0.00	0.00	0.00	0.00
113	33.48	801.88	0.40	0.00	0.00	0.00	0.00
114	27.96	761.47	-11.38	0.00	0.00	0.00	0.00
115	-141.53	730.13	-34.83	0.00	0.00	0.00	0.00
116	-146.03	810.45	15.58	0.00	0.00	0.00	0.00
117	200.22	789.55	-44.35	0.00	0.00	0.00	0.00
118	195.72	869.87	6.05	0.00	0.00	0.00	0.00
119	-17.74	659.49	-95.41	0.00	0.00	0.00	0.00
120	86.68	677.65	-98.33	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
121	-32.49	922.35	69.55	0.00	0.00	0.00	0.00
122	71.93	940.51	66.64	0.00	0.00	0.00	0.00
123	-144.23	650.13	-33.39	0.00	0.00	0.00	0.00
124	-148.74	730.45	17.02	0.00	0.00	0.00	0.00
125	197.51	709.55	-42.92	0.00	0.00	0.00	0.00
126	193.01	789.87	7.49	0.00	0.00	0.00	0.00
127	-20.45	579.49	-93.98	0.00	0.00	0.00	0.00
128	83.97	597.65	-96.89	0.00	0.00	0.00	0.00
129	-35.20	842.35	70.99	0.00	0.00	0.00	0.00
130	69.22	860.51	68.08	0.00	0.00	0.00	0.00
131	-88.90	927.35	-21.38	0.00	0.00	0.00	0.00
132	-92.38	989.42	17.57	0.00	0.00	0.00	0.00
133	167.41	971.92	-28.52	0.00	0.00	0.00	0.00
134	163.93	1033.98	10.43	0.00	0.00	0.00	0.00
135	2.70	875.08	-66.21	0.00	0.00	0.00	0.00
136	83.39	889.11	-68.46	0.00	0.00	0.00	0.00
137	-8.36	1072.22	57.51	0.00	0.00	0.00	0.00
138	72.33	1086.25	55.26	0.00	0.00	0.00	0.00
101	-72.66	586.46	46.12	0.00	0.00	0.00	0.00
102	-60.47	521.89	38.92	0.00	0.00	0.00	0.00
103	-58.83	485.70	39.75	0.00	0.00	0.00	0.00
104	-49.15	478.00	35.27	0.00	0.00	0.00	0.00
105	-58.76	472.16	39.18	0.00	0.00	0.00	0.00
106	-49.07	461.07	34.57	0.00	0.00	0.00	0.00
107	-58.04	477.65	37.16	0.00	0.00	0.00	0.00
108	-48.17	467.93	32.04	0.00	0.00	0.00	0.00
109	-58.27	452.89	37.98	0.00	0.00	0.00	0.00
110	-48.45	436.98	33.07	0.00	0.00	0.00	0.00
111	-57.44	453.50	33.33	0.00	0.00	0.00	0.00
112	-47.42	437.75	27.26	0.00	0.00	0.00	0.00
113	-57.72	441.95	35.13	0.00	0.00	0.00	0.00
114	-47.77	423.31	29.50	0.00	0.00	0.00	0.00
115	-139.83	542.53	10.68	0.00	0.00	0.00	0.00
116	-142.60	629.29	47.28	0.00	0.00	0.00	0.00
117	46.04	285.35	16.15	0.00	0.00	0.00	0.00
118	43.26	372.12	52.75	0.00	0.00	0.00	0.00
119	-72.14	354.63	-29.01	0.00	0.00	0.00	0.00
120	-15.35	276.05	-27.34	0.00	0.00	0.00	0.00
121	-81.22	638.60	90.77	0.00	0.00	0.00	0.00
122	-24.42	560.02	92.44	0.00	0.00	0.00	0.00
123	-135.00	496.79	7.51	0.00	0.00	0.00	0.00
124	-137.77	583.56	44.11	0.00	0.00	0.00	0.00
125	50.87	239.62	12.98	0.00	0.00	0.00	0.00
126	48.09	326.39	49.58	0.00	0.00	0.00	0.00
127	-67.31	308.90	-32.18	0.00	0.00	0.00	0.00
128	-10.52	230.32	-30.51	0.00	0.00	0.00	0.00
129	-76.39	592.87	87.60	0.00	0.00	0.00	0.00
130	-19.60	514.29	89.27	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
131	-135.19	617.09	26.33	0.00	0.00	0.00	0.00
132	-137.34	684.14	54.61	0.00	0.00	0.00	0.00
133	4.20	424.21	30.43	0.00	0.00	0.00	0.00
134	2.06	491.26	58.71	0.00	0.00	0.00	0.00
135	-85.10	478.05	-3.04	0.00	0.00	0.00	0.00
136	-41.22	417.32	-1.75	0.00	0.00	0.00	0.00
137	-91.91	691.02	86.79	0.00	0.00	0.00	0.00
138	-48.03	630.30	88.08	0.00	0.00	0.00	0.00
5	101	37.44	1012.38	18.58	0.00	0.00	0.00
102	22.33	878.54	14.36	0.00	0.00	0.00	0.00
103	31.23	779.24	29.70	0.00	0.00	0.00	0.00
104	8.81	706.39	28.69	0.00	0.00	0.00	0.00
105	32.06	757.46	26.14	0.00	0.00	0.00	0.00
106	9.86	679.16	24.24	0.00	0.00	0.00	0.00
107	29.04	832.87	14.73	0.00	0.00	0.00	0.00
108	6.08	773.43	9.98	0.00	0.00	0.00	0.00
109	31.50	770.20	15.25	0.00	0.00	0.00	0.00
110	9.16	695.08	10.63	0.00	0.00	0.00	0.00
111	28.69	840.70	-0.04	0.00	0.00	0.00	0.00
112	5.64	783.21	-8.49	0.00	0.00	0.00	0.00
113	30.29	800.22	4.07	0.00	0.00	0.00	0.00
114	7.65	732.62	-3.35	0.00	0.00	0.00	0.00
115	5.59	687.72	-32.36	0.00	0.00	0.00	0.00
116	7.11	640.20	60.67	0.00	0.00	0.00	0.00
117	7.34	849.22	-40.38	0.00	0.00	0.00	0.00
118	8.85	801.70	52.66	0.00	0.00	0.00	0.00
119	4.48	797.81	-140.87	0.00	0.00	0.00	0.00
120	5.01	847.15	-143.32	0.00	0.00	0.00	0.00
121	9.44	642.27	163.62	0.00	0.00	0.00	0.00
122	9.97	691.62	161.17	0.00	0.00	0.00	0.00
123	4.87	613.25	-33.38	0.00	0.00	0.00	0.00
124	6.39	565.73	59.66	0.00	0.00	0.00	0.00
125	6.62	774.75	-41.40	0.00	0.00	0.00	0.00
126	8.13	727.23	51.64	0.00	0.00	0.00	0.00
127	3.76	723.33	-141.89	0.00	0.00	0.00	0.00
128	4.29	772.68	-144.34	0.00	0.00	0.00	0.00
129	8.72	567.80	162.60	0.00	0.00	0.00	0.00
130	9.25	617.15	160.15	0.00	0.00	0.00	0.00
131	28.65	903.26	-16.47	0.00	0.00	0.00	0.00
132	29.82	866.53	55.43	0.00	0.00	0.00	0.00
133	29.96	1024.38	-22.48	0.00	0.00	0.00	0.00
134	31.13	987.66	49.41	0.00	0.00	0.00	0.00
135	27.82	984.72	-96.76	0.00	0.00	0.00	0.00
136	28.23	1022.85	-98.66	0.00	0.00	0.00	0.00
137	31.54	868.07	131.60	0.00	0.00	0.00	0.00
138	31.95	906.20	129.71	0.00	0.00	0.00	0.00
6	101	-14.20	719.28	26.87	0.00	0.00	0.00
102	-10.37	526.82	20.38	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
103	-11.32	569.39	40.51	0.00	0.00	0.00	0.00
104	-6.49	326.83	37.65	0.00	0.00	0.00	0.00
105	-11.34	571.64	36.02	0.00	0.00	0.00	0.00
106	-6.52	329.64	32.03	0.00	0.00	0.00	0.00
107	-11.34	574.63	21.69	0.00	0.00	0.00	0.00
108	-6.51	333.38	14.13	0.00	0.00	0.00	0.00
109	-11.38	576.67	21.30	0.00	0.00	0.00	0.00
110	-6.56	335.93	13.64	0.00	0.00	0.00	0.00
111	-11.40	581.50	2.44	0.00	0.00	0.00	0.00
112	-6.59	341.96	-9.94	0.00	0.00	0.00	0.00
113	-11.41	581.24	6.58	0.00	0.00	0.00	0.00
114	-6.60	341.64	-4.76	0.00	0.00	0.00	0.00
115	-103.48	357.14	-38.42	0.00	0.00	0.00	0.00
116	-103.49	320.91	65.98	0.00	0.00	0.00	0.00
117	90.42	347.82	-38.21	0.00	0.00	0.00	0.00
118	90.40	311.60	66.19	0.00	0.00	0.00	0.00
119	-36.14	395.07	-156.98	0.00	0.00	0.00	0.00
120	23.11	392.22	-156.91	0.00	0.00	0.00	0.00
121	-36.18	276.51	184.69	0.00	0.00	0.00	0.00
122	23.07	273.67	184.75	0.00	0.00	0.00	0.00
123	-102.82	323.70	-39.81	0.00	0.00	0.00	0.00
124	-102.83	287.48	64.59	0.00	0.00	0.00	0.00
125	91.07	314.38	-39.60	0.00	0.00	0.00	0.00
126	91.06	278.16	64.80	0.00	0.00	0.00	0.00
127	-35.48	361.63	-158.37	0.00	0.00	0.00	0.00
128	23.76	358.78	-158.30	0.00	0.00	0.00	0.00
129	-35.52	243.08	183.30	0.00	0.00	0.00	0.00
130	23.72	240.23	183.36	0.00	0.00	0.00	0.00
131	-84.99	640.54	-16.79	0.00	0.00	0.00	0.00
132	-85.00	612.55	63.88	0.00	0.00	0.00	0.00
133	60.43	633.55	-16.63	0.00	0.00	0.00	0.00
134	60.42	605.56	64.04	0.00	0.00	0.00	0.00
135	-35.16	668.61	-104.52	0.00	0.00	0.00	0.00
136	10.62	666.41	-104.47	0.00	0.00	0.00	0.00
137	-35.19	579.69	151.73	0.00	0.00	0.00	0.00
138	10.59	577.49	151.78	0.00	0.00	0.00	0.00
7	101	28.86	709.83	38.37	0.00	0.00	0.00
102	19.76	516.68	29.59	0.00	0.00	0.00	0.00
103	23.65	562.14	56.37	0.00	0.00	0.00	0.00
104	11.35	316.38	52.91	0.00	0.00	0.00	0.00
105	23.91	564.24	50.55	0.00	0.00	0.00	0.00
106	11.69	319.00	45.63	0.00	0.00	0.00	0.00
107	22.78	567.03	31.06	0.00	0.00	0.00	0.00
108	10.27	322.50	21.28	0.00	0.00	0.00	0.00
109	23.62	569.01	30.45	0.00	0.00	0.00	0.00
110	11.32	324.97	20.51	0.00	0.00	0.00	0.00
111	22.52	573.70	4.88	0.00	0.00	0.00	0.00
112	9.94	330.84	-11.45	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
113	23.09	573.50	10.19	0.00	0.00	0.00	0.00
114	10.66	330.58	-4.81	0.00	0.00	0.00	0.00
115	-168.97	315.27	-44.32	0.00	0.00	0.00	0.00
116	-168.19	282.08	89.74	0.00	0.00	0.00	0.00
117	189.50	365.00	-48.11	0.00	0.00	0.00	0.00
118	190.28	331.80	85.96	0.00	0.00	0.00	0.00
119	-45.39	370.25	-197.98	0.00	0.00	0.00	0.00
120	64.14	385.45	-199.14	0.00	0.00	0.00	0.00
121	-42.84	261.63	240.77	0.00	0.00	0.00	0.00
122	66.69	276.82	239.62	0.00	0.00	0.00	0.00
123	-170.04	282.92	-46.40	0.00	0.00	0.00	0.00
124	-169.26	249.72	87.66	0.00	0.00	0.00	0.00
125	188.43	332.64	-50.19	0.00	0.00	0.00	0.00
126	189.21	299.45	83.87	0.00	0.00	0.00	0.00
127	-46.45	337.90	-200.06	0.00	0.00	0.00	0.00
128	63.08	353.09	-201.22	0.00	0.00	0.00	0.00
129	-43.90	229.27	238.69	0.00	0.00	0.00	0.00
130	65.63	244.47	237.53	0.00	0.00	0.00	0.00
131	-110.42	607.43	-16.40	0.00	0.00	0.00	0.00
132	-109.82	581.78	87.20	0.00	0.00	0.00	0.00
133	158.43	644.73	-19.24	0.00	0.00	0.00	0.00
134	159.04	619.08	84.36	0.00	0.00	0.00	0.00
135	-18.97	648.12	-130.11	0.00	0.00	0.00	0.00
136	65.67	659.86	-131.00	0.00	0.00	0.00	0.00
137	-17.05	566.65	198.96	0.00	0.00	0.00	0.00
138	67.58	578.39	198.07	0.00	0.00	0.00	0.00
8	101	-43.82	1008.24	27.79	0.00	0.00	0.00
102	-24.73	870.40	22.40	0.00	0.00	0.00	0.00
103	-36.04	776.89	32.89	0.00	0.00	0.00	0.00
104	-6.87	695.42	30.34	0.00	0.00	0.00	0.00
105	-37.14	752.99	30.63	0.00	0.00	0.00	0.00
106	-8.25	665.55	27.51	0.00	0.00	0.00	0.00
107	-33.99	830.70	22.26	0.00	0.00	0.00	0.00
108	-4.31	762.68	17.05	0.00	0.00	0.00	0.00
109	-36.90	764.33	22.97	0.00	0.00	0.00	0.00
110	-7.95	679.73	17.93	0.00	0.00	0.00	0.00
111	-34.07	836.74	11.22	0.00	0.00	0.00	0.00
112	-4.41	770.23	3.25	0.00	0.00	0.00	0.00
113	-35.86	794.54	14.39	0.00	0.00	0.00	0.00
114	-6.65	717.49	7.22	0.00	0.00	0.00	0.00
115	-41.95	852.50	-20.74	0.00	0.00	0.00	0.00
116	-42.18	810.38	46.49	0.00	0.00	0.00	0.00
117	30.89	654.73	-12.46	0.00	0.00	0.00	0.00
118	30.66	612.61	54.77	0.00	0.00	0.00	0.00
119	-16.40	831.69	-94.25	0.00	0.00	0.00	0.00
120	5.86	771.26	-91.72	0.00	0.00	0.00	0.00
121	-17.15	693.85	125.76	0.00	0.00	0.00	0.00
122	5.11	633.42	128.29	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
123	-41.38	779.24	-22.44	0.00	0.00	0.00	0.00
124	-41.61	737.13	44.79	0.00	0.00	0.00	0.00
125	31.45	581.48	-14.16	0.00	0.00	0.00	0.00
126	31.22	539.36	53.07	0.00	0.00	0.00	0.00
127	-15.83	758.43	-95.96	0.00	0.00	0.00	0.00
128	6.42	698.01	-93.43	0.00	0.00	0.00	0.00
129	-16.58	620.60	124.06	0.00	0.00	0.00	0.00
130	5.67	560.17	126.59	0.00	0.00	0.00	0.00
131	-61.50	1029.76	-3.98	0.00	0.00	0.00	0.00
132	-61.68	997.21	47.96	0.00	0.00	0.00	0.00
133	-6.87	881.43	2.23	0.00	0.00	0.00	0.00
134	-7.05	848.89	54.18	0.00	0.00	0.00	0.00
135	-42.59	1014.36	-58.39	0.00	0.00	0.00	0.00
136	-25.40	967.66	-56.43	0.00	0.00	0.00	0.00
137	-43.16	910.98	106.62	0.00	0.00	0.00	0.00
138	-25.96	864.29	108.58	0.00	0.00	0.00	0.00
9	101	25.16	870.72	-19.10	0.00	0.00	0.00
102	17.74	758.30	-14.64	0.00	0.00	0.00	0.00
103	20.09	724.92	-8.71	0.00	0.00	0.00	0.00
104	10.28	681.32	-1.97	0.00	0.00	0.00	0.00
105	20.14	688.56	-10.44	0.00	0.00	0.00	0.00
106	10.34	635.87	-4.14	0.00	0.00	0.00	0.00
107	20.10	716.97	-15.44	0.00	0.00	0.00	0.00
108	10.29	671.38	-10.39	0.00	0.00	0.00	0.00
109	20.18	660.85	-15.33	0.00	0.00	0.00	0.00
110	10.39	601.23	-10.25	0.00	0.00	0.00	0.00
111	20.16	668.57	-21.69	0.00	0.00	0.00	0.00
112	10.37	610.89	-18.21	0.00	0.00	0.00	0.00
113	20.19	649.05	-19.94	0.00	0.00	0.00	0.00
114	10.41	586.49	-16.01	0.00	0.00	0.00	0.00
115	-14.74	500.61	-30.26	0.00	0.00	0.00	0.00
116	-15.23	544.48	13.87	0.00	0.00	0.00	0.00
117	35.88	747.29	-34.25	0.00	0.00	0.00	0.00
118	35.39	791.16	9.88	0.00	0.00	0.00	0.00
119	3.39	536.42	-81.79	0.00	0.00	0.00	0.00
120	18.86	611.79	-83.01	0.00	0.00	0.00	0.00
121	1.80	679.99	62.64	0.00	0.00	0.00	0.00
122	17.26	755.36	61.41	0.00	0.00	0.00	0.00
123	-15.77	436.03	-29.24	0.00	0.00	0.00	0.00
124	-16.26	479.89	14.89	0.00	0.00	0.00	0.00
125	34.85	682.71	-33.23	0.00	0.00	0.00	0.00
126	34.36	726.57	10.90	0.00	0.00	0.00	0.00
127	2.36	471.83	-80.77	0.00	0.00	0.00	0.00
128	17.82	547.20	-81.99	0.00	0.00	0.00	0.00
129	0.77	615.40	63.65	0.00	0.00	0.00	0.00
130	16.23	690.77	62.43	0.00	0.00	0.00	0.00
131	2.66	705.06	-32.42	0.00	0.00	0.00	0.00
132	2.28	738.95	1.68	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
10	133	40.62	890.07	-35.42	0.00	0.00	0.00
	134	40.24	923.97	-1.32	0.00	0.00	0.00
	135	16.07	731.55	-70.56	0.00	0.00	0.00
	136	28.02	789.79	-71.50	0.00	0.00	0.00
	137	14.88	839.23	37.76	0.00	0.00	0.00
	138	26.83	897.47	36.82	0.00	0.00	0.00
	101	18.34	1440.01	3.30	0.00	0.00	0.00
	102	17.14	1240.11	9.07	0.00	0.00	0.00
	103	14.61	1182.02	4.35	0.00	0.00	0.00
	104	15.87	1077.72	16.97	0.00	0.00	0.00
11	105	14.66	1144.02	2.66	0.00	0.00	0.00
	106	15.92	1030.23	14.86	0.00	0.00	0.00
	107	14.65	1175.05	3.19	0.00	0.00	0.00
	108	15.92	1069.01	15.52	0.00	0.00	0.00
	109	14.71	1112.33	1.36	0.00	0.00	0.00
	110	15.99	990.61	13.24	0.00	0.00	0.00
	111	14.72	1122.16	1.10	0.00	0.00	0.00
	112	16.00	1002.90	12.91	0.00	0.00	0.00
	113	14.74	1097.84	0.89	0.00	0.00	0.00
	114	16.03	972.50	12.65	0.00	0.00	0.00
12	115	-39.47	1021.58	-1.33	0.00	0.00	0.00
	116	-40.08	1091.18	31.89	0.00	0.00	0.00
	117	71.96	989.24	-2.21	0.00	0.00	0.00
	118	71.35	1058.85	31.01	0.00	0.00	0.00
	119	-0.10	931.26	-39.39	0.00	0.00	0.00
	120	33.95	921.38	-39.66	0.00	0.00	0.00
	121	-2.07	1159.05	69.34	0.00	0.00	0.00
	122	31.98	1149.17	69.07	0.00	0.00	0.00
	123	-41.07	917.56	-2.82	0.00	0.00	0.00
	124	-41.67	987.16	30.41	0.00	0.00	0.00
13	125	70.36	885.22	-3.70	0.00	0.00	0.00
	126	69.76	954.83	29.53	0.00	0.00	0.00
	127	-1.69	827.23	-40.88	0.00	0.00	0.00
	128	32.36	817.35	-41.14	0.00	0.00	0.00
	129	-3.67	1055.03	67.86	0.00	0.00	0.00
	130	30.38	1045.15	67.59	0.00	0.00	0.00
	131	-23.82	1325.30	-6.32	0.00	0.00	0.00
	132	-24.28	1379.08	19.35	0.00	0.00	0.00
	133	59.76	1301.04	-6.98	0.00	0.00	0.00
	134	59.29	1354.83	18.69	0.00	0.00	0.00
14	135	5.32	1258.46	-34.48	0.00	0.00	0.00
	136	33.61	1250.82	-34.69	0.00	0.00	0.00
	137	3.84	1429.30	47.06	0.00	0.00	0.00
	138	30.15	1421.67	46.86	0.00	0.00	0.00
	139	22.53	1624.34	16.04	0.00	0.00	0.00
	140	18.48	1405.34	22.04	0.00	0.00	0.00
	141	17.85	1329.24	13.91	0.00	0.00	0.00
	142	14.22	1223.54	29.38	0.00	0.00	0.00
	143	14.22	1223.54	29.38	0.00	0.00	0.00
	144	14.22	1223.54	29.38	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
10	105	17.91	1287.20	11.65	0.00	0.00	0.00
	106	14.28	1170.99	26.56	0.00	0.00	0.00
	107	18.02	1325.38	13.73	0.00	0.00	0.00
	108	14.42	1218.72	29.16	0.00	0.00	0.00
	109	18.04	1254.51	10.63	0.00	0.00	0.00
	110	14.45	1130.13	25.29	0.00	0.00	0.00
	111	18.18	1270.03	12.08	0.00	0.00	0.00
	112	14.62	1149.53	27.09	0.00	0.00	0.00
	113	18.15	1241.59	11.08	0.00	0.00	0.00
	114	14.59	1113.97	25.84	0.00	0.00	0.00
11	115	-120.22	1105.28	9.68	0.00	0.00	0.00
	116	-121.64	1175.58	48.32	0.00	0.00	0.00
	117	150.49	1197.09	7.75	0.00	0.00	0.00
	118	149.07	1267.38	46.39	0.00	0.00	0.00
	119	-24.61	1057.28	-34.89	0.00	0.00	0.00
	120	58.11	1085.34	-35.48	0.00	0.00	0.00
	121	-29.25	1287.33	91.56	0.00	0.00	0.00
	122	53.46	1315.38	90.97	0.00	0.00	0.00
	123	-121.67	986.65	6.88	0.00	0.00	0.00
	124	-123.08	1056.94	45.52	0.00	0.00	0.00
12	125	149.05	1078.45	4.95	0.00	0.00	0.00
	126	147.63	1148.75	43.59	0.00	0.00	0.00
	127	-26.05	938.65	-37.70	0.00	0.00	0.00
	128	56.66	966.70	-38.29	0.00	0.00	0.00
	129	-30.70	1168.69	88.75	0.00	0.00	0.00
	130	52.02	1196.75	88.16	0.00	0.00	0.00
	131	-80.47	1453.25	4.84	0.00	0.00	0.00
	132	-81.56	1507.57	34.69	0.00	0.00	0.00
	133	122.57	1522.11	3.39	0.00	0.00	0.00
	134	121.47	1576.42	33.25	0.00	0.00	0.00
13	135	-9.72	1417.73	-28.15	0.00	0.00	0.00
	136	54.20	1439.41	-28.61	0.00	0.00	0.00
	137	-13.20	1590.27	66.69	0.00	0.00	0.00
	138	50.72	1611.94	66.23	0.00	0.00	0.00
	139	-74.85	1018.63	-24.20	0.00	0.00	0.00
	140	-60.74	890.40	-19.70	0.00	0.00	0.00
	141	-60.16	844.08	-8.00	0.00	0.00	0.00
	142	-46.98	798.63	-1.01	0.00	0.00	0.00
	143	-60.12	806.43	-10.77	0.00	0.00	0.00
	144	-46.94	751.57	-4.47	0.00	0.00	0.00
14	145	-59.83	836.03	-19.76	0.00	0.00	0.00
	146	-46.58	788.57	-15.70	0.00	0.00	0.00
	147	-59.91	777.69	-19.50	0.00	0.00	0.00
	148	-46.68	715.63	-15.38	0.00	0.00	0.00
	149	-59.62	786.20	-30.33	0.00	0.00	0.00
	150	-46.31	726.27	-28.91	0.00	0.00	0.00
	151	-59.72	765.69	-27.52	0.00	0.00	0.00
	152	-46.44	700.64	-25.40	0.00	0.00	0.00
	153	-46.44	700.64	-25.40	0.00	0.00	0.00
	154	-46.44	700.64	-25.40	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
115	-219.23	892.66	-55.13	0.00	0.00	0.00	0.00
116	-221.52	939.57	14.02	0.00	0.00	0.00	0.00
117	128.25	584.74	-44.43	0.00	0.00	0.00	0.00
118	125.96	631.65	24.72	0.00	0.00	0.00	0.00
119	-95.97	732.44	-130.00	0.00	0.00	0.00	0.00
120	10.20	638.35	-126.73	0.00	0.00	0.00	0.00
121	-103.47	885.96	96.32	0.00	0.00	0.00	0.00
122	2.71	791.88	99.59	0.00	0.00	0.00	0.00
123	-214.56	816.45	-53.61	0.00	0.00	0.00	0.00
124	-216.85	863.36	15.54	0.00	0.00	0.00	0.00
125	132.92	508.53	-42.91	0.00	0.00	0.00	0.00
126	130.63	555.44	26.25	0.00	0.00	0.00	0.00
127	-91.31	656.22	-128.47	0.00	0.00	0.00	0.00
128	14.86	562.14	-125.20	0.00	0.00	0.00	0.00
129	-98.80	809.75	97.84	0.00	0.00	0.00	0.00
130	7.37	715.66	101.11	0.00	0.00	0.00	0.00
131	-197.21	1051.86	-52.68	0.00	0.00	0.00	0.00
132	-198.98	1088.11	0.75	0.00	0.00	0.00	0.00
133	63.40	820.92	-44.66	0.00	0.00	0.00	0.00
134	61.63	857.17	8.78	0.00	0.00	0.00	0.00
135	-106.00	933.29	-108.08	0.00	0.00	0.00	0.00
136	-23.96	860.59	-105.56	0.00	0.00	0.00	0.00
137	-111.62	1048.44	61.65	0.00	0.00	0.00	0.00
138	-29.58	975.73	64.18	0.00	0.00	0.00	0.00
13	101	25.23	475.87	-15.22	0.00	0.00	0.00
	102	23.44	449.72	-13.83	0.00	0.00	0.00
	103	20.35	363.08	-9.28	0.00	0.00	0.00
	104	21.85	401.55	-8.81	0.00	0.00	0.00
	105	20.30	350.77	-10.29	0.00	0.00	0.00
	106	21.79	386.16	-10.08	0.00	0.00	0.00
	107	20.18	390.05	-12.39	0.00	0.00	0.00
	108	21.64	435.26	-12.70	0.00	0.00	0.00
	109	20.17	362.83	-12.37	0.00	0.00	0.00
	110	21.63	401.24	-12.68	0.00	0.00	0.00
	111	20.03	399.23	-14.82	0.00	0.00	0.00
	112	21.45	446.74	-15.74	0.00	0.00	0.00
	113	20.06	383.98	-13.93	0.00	0.00	0.00
114	21.50	427.67	-14.64	0.00	0.00	0.00	0.00
115	-2.77	369.52	-24.88	0.00	0.00	0.00	0.00
116	-1.71	274.55	2.89	0.00	0.00	0.00	0.00
117	45.00	572.59	-27.77	0.00	0.00	0.00	0.00
118	46.06	477.62	0.00	0.00	0.00	0.00	0.00
119	12.62	547.94	-57.44	0.00	0.00	0.00	0.00
120	27.22	609.99	-58.32	0.00	0.00	0.00	0.00
121	16.07	237.14	33.45	0.00	0.00	0.00	0.00
122	30.67	299.19	32.56	0.00	0.00	0.00	0.00
123	-4.93	327.16	-23.64	0.00	0.00	0.00	0.00
124	-3.88	232.19	4.13	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
125	42.84	530.23	-26.53	0.00	0.00	0.00	0.00
126	43.89	435.26	1.25	0.00	0.00	0.00	0.00
127	10.46	505.59	-56.20	0.00	0.00	0.00	0.00
128	25.05	567.64	-57.08	0.00	0.00	0.00	0.00
129	13.91	194.79	34.69	0.00	0.00	0.00	0.00
130	28.50	256.84	33.81	0.00	0.00	0.00	0.00
131	6.01	423.33	-24.17	0.00	0.00	0.00	0.00
132	6.83	349.95	-2.71	0.00	0.00	0.00	0.00
133	41.84	575.64	-26.34	0.00	0.00	0.00	0.00
134	42.65	502.25	-4.88	0.00	0.00	0.00	0.00
135	17.40	555.37	-48.27	0.00	0.00	0.00	0.00
136	28.68	603.32	-48.95	0.00	0.00	0.00	0.00
137	19.99	322.27	19.90	0.00	0.00	0.00	0.00
138	31.27	370.22	19.22	0.00	0.00	0.00	0.00
14	101	11.84	789.28	-20.46	0.00	0.00	0.00
	102	8.26	722.46	-17.62	0.00	0.00	0.00
	103	9.74	609.90	-17.11	0.00	0.00	0.00
	104	5.01	628.74	-15.71	0.00	0.00	0.00
	105	9.66	596.76	-17.63	0.00	0.00	0.00
	106	4.92	612.31	-16.35	0.00	0.00	0.00
	107	9.47	641.16	-16.51	0.00	0.00	0.00
	108	4.67	667.81	-14.94	0.00	0.00	0.00
	109	9.46	612.59	-16.78	0.00	0.00	0.00
	110	4.66	632.10	-15.29	0.00	0.00	0.00
	111	9.21	653.82	-15.32	0.00	0.00	0.00
	112	4.35	683.63	-13.46	0.00	0.00	0.00
	113	9.27	638.02	-15.25	0.00	0.00	0.00
114	4.43	663.88	-13.38	0.00	0.00	0.00	0.00
115	-45.24	710.26	-24.61	0.00	0.00	0.00	0.00
116	-43.53	595.45	-5.95	0.00	0.00	0.00	0.00
117	52.88	715.82	-23.60	0.00	0.00	0.00	0.00
118	54.60	601.02	-4.94	0.00	0.00	0.00	0.00
119	-13.12	842.65	-45.46	0.00	0.00	0.00	0.00
120	16.86	844.35	-45.15	0.00	0.00	0.00	0.00
121	-7.51	466.92	15.61	0.00	0.00	0.00	0.00
122	22.48	468.62	15.92	0.00	0.00	0.00	0.00
123	-45.71	644.69	-23.13	0.00	0.00	0.00	0.00
124	-43.99	529.89	-4.47	0.00	0.00	0.00	0.00
125	52.41	650.26	-22.12	0.00	0.00	0.00	0.00
126	54.13	535.45	-3.46	0.00	0.00	0.00	0.00
127	-13.59	777.09	-43.99	0.00	0.00	0.00	0.00
128	16.39	778.79	-43.68	0.00	0.00	0.00	0.00
129	-7.97	401.36	17.08	0.00	0.00	0.00	0.00
130	22.01	403.06	17.39	0.00	0.00	0.00	0.00
131	-27.41	798.14	-26.63	0.00	0.00	0.00	0.00
132	-26.08	709.42	-12.21	0.00	0.00	0.00	0.00
133	46.18	802.31	-25.87	0.00	0.00	0.00	0.00
134	47.51	713.60	-11.45	0.00	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN										METE		STRUCTURE TYPE = SPACE			
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JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z	JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
15	135	-3.64	896.11	-42.06	0.00	0.00	0.00	16	107	-36.35	462.45	-45.12	0.00	0.00	0.00
	136	19.53	897.42	-41.82	0.00	0.00	0.00		108	-37.14	470.98	-38.84	0.00	0.00	0.00
	137	0.57	614.31	3.74	0.00	0.00	0.00		109	-36.41	435.44	-45.23	0.00	0.00	0.00
	138	23.74	615.63	3.98	0.00	0.00	0.00		110	-37.22	437.23	-38.97	0.00	0.00	0.00
	101	11.17	899.15	-47.76	0.00	0.00	0.00		111	-37.19	472.03	-47.81	0.00	0.00	0.00
	102	11.79	801.82	-39.11	0.00	0.00	0.00		112	-38.20	482.96	-42.19	0.00	0.00	0.00
	103	9.56	698.19	-40.69	0.00	0.00	0.00		113	-37.02	456.88	-46.82	0.00	0.00	0.00
	104	13.19	678.09	-33.57	0.00	0.00	0.00		114	-37.98	464.03	-40.96	0.00	0.00	0.00
	105	9.38	685.43	-41.13	0.00	0.00	0.00		115	-190.99	638.94	-60.20	0.00	0.00	0.00
	106	12.97	662.14	-34.11	0.00	0.00	0.00		116	-185.41	544.24	-21.26	0.00	0.00	0.00
	107	8.94	728.75	-38.46	0.00	0.00	0.00		117	111.09	374.38	-55.07	0.00	0.00	0.00
	108	12.42	716.29	-30.77	0.00	0.00	0.00		118	116.67	279.68	-16.13	0.00	0.00	0.00
	109	8.90	701.17	-38.98	0.00	0.00	0.00		119	-92.45	654.70	-102.67	0.00	0.00	0.00
	110	12.38	681.81	-31.43	0.00	0.00	0.00		120	-0.15	573.86	-101.10	0.00	0.00	0.00
16	111	8.32	741.24	-35.16	0.00	0.00	0.00		121	-74.17	344.76	24.76	0.00	0.00	0.00
	112	11.65	731.90	-26.65	0.00	0.00	0.00		122	18.13	263.93	26.33	0.00	0.00	0.00
	113	8.45	726.07	-35.57	0.00	0.00	0.00		123	-187.28	593.01	-56.39	0.00	0.00	0.00
	114	11.82	712.94	-27.16	0.00	0.00	0.00		124	-181.69	498.31	-17.45	0.00	0.00	0.00
	115	-107.69	733.01	-38.14	0.00	0.00	0.00		125	114.80	328.45	-51.25	0.00	0.00	0.00
	116	-103.65	618.88	-20.65	0.00	0.00	0.00		126	120.39	233.75	-12.32	0.00	0.00	0.00
	117	128.48	790.11	-40.27	0.00	0.00	0.00		127	-88.73	608.77	-98.85	0.00	0.00	0.00
	118	132.52	675.98	-22.79	0.00	0.00	0.00		128	3.57	527.93	-97.28	0.00	0.00	0.00
	119	-30.28	882.53	-58.74	0.00	0.00	0.00		129	-70.46	298.83	28.58	0.00	0.00	0.00
	120	41.88	899.98	-59.40	0.00	0.00	0.00		130	21.85	218.00	30.15	0.00	0.00	0.00
	121	-17.05	509.01	-1.53	0.00	0.00	0.00		131	-158.82	675.42	-68.31	0.00	0.00	0.00
	122	55.11	526.46	-2.18	0.00	0.00	0.00		132	-154.50	602.24	-38.23	0.00	0.00	0.00
	123	-108.93	662.56	-35.09	0.00	0.00	0.00		133	67.74	477.00	-64.46	0.00	0.00	0.00
	124	-104.89	548.43	-17.61	0.00	0.00	0.00		134	72.06	403.82	-34.38	0.00	0.00	0.00
	125	127.24	719.66	-37.23	0.00	0.00	0.00		135	-85.90	687.08	-99.74	0.00	0.00	0.00
	126	131.28	605.53	-19.74	0.00	0.00	0.00		136	-14.57	624.61	-98.53	0.00	0.00	0.00
	127	-31.52	812.08	-55.70	0.00	0.00	0.00		137	-72.19	454.63	-4.16	0.00	0.00	0.00
	128	40.64	829.53	-56.35	0.00	0.00	0.00		138	-0.86	392.16	-2.95	0.00	0.00	0.00
	129	-18.29	438.56	1.52	0.00	0.00	0.00	***** END OF LATEST ANALYSIS RESULT *****							
130	53.87	456.01	0.86	0.00	0.00	0.00	743. *****								
131	-78.65	873.17	-49.39	0.00	0.00	0.00	744. LOAD LIST 201 TO 236								
132	-75.53	784.97	-35.88	0.00	0.00	0.00	745. START CONCRETE DESIGN								
133	98.48	915.99	-50.99	0.00	0.00	0.00	[CONCRETE DESIGN]								
134	101.60	827.80	-37.48	0.00	0.00	0.00	746. CODE BS8110								
135	-21.37	983.81	-64.64	0.00	0.00	0.00	PROGRAM CODE REVISION V1.5_8110_97/1								
136	34.40	997.30	-65.14	0.00	0.00	0.00	747. UNIT MMS NEWTON								
137	-11.44	703.67	-21.73	0.00	0.00	0.00	748. FYMAIN 460 ALL								
138	44.32	717.16	-22.23	0.00	0.00	0.00									
101	-45.45	566.39	-55.74	0.00	0.00	0.00									
102	-41.31	512.85	-46.95	0.00	0.00	0.00									
103	-35.52	435.04	-40.64	0.00	0.00	0.00									
104	-36.11	436.72	-33.23	0.00	0.00	0.00									
105	-35.77	422.88	-42.15	0.00	0.00	0.00									
106	-36.42	421.53	-35.12	0.00	0.00	0.00									

C O L U M N N O. 1001 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2726. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
36 10 MM	0.935	225	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

749. FC 40 ALL

750. MINSEC 10 ALL

IBRACED /SHORT in z E.L.z= 3900 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 3900 mm (3.8.1.3 & 5) |
END MOMS. MZ1= 0 MZ2= 157 MY1= 0 MY2= 298 |
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 442 AXIAL LOAD= 881 |
DESIGNED CAP. KN METER: MOM.= 442 AXIAL CAP.= 882 |

751. CLEAR 40 MEMB 1 TO 42 101 TO 124

C O L U M N N O. 1002 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 3522. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
32 12 MM	1.196	231	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

753. TRACK 2 ALL

754. DESIGN COLUMN 1001 TO 1016 2001 TO 2005 2008 TO 2016

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C O L U M N N O. 1003 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 5759. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

12 25 MM 1.947 229 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 2000 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 2000 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 582 MY1= 0 MY2= 118|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 692 AXIAL LOAD= 777|
DESIGNED CAP. KN METER: MOM.= 692 AXIAL CAP.= 784

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C O L U M N N O. 1004 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 4424. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

40 12 MM 1.495 220 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 2000 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 2000 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 469 MY1= 0 MY2= 161|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 615 AXIAL LOAD= 1007|
DESIGNED CAP. KN METER: MOM.= 615 AXIAL CAP.= 1015

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C O L U M N N O. 1005 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 6363. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

32 16 MM 2.127 225 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 2700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 2700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 124 MY1= 0 MY2= 694|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 805 AXIAL LOAD= 1121|
DESIGNED CAP. KN METER: MOM.= 805 AXIAL CAP.= 1123

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C O L U M N N O. 1006 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 8985. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

12 32 MM 3.191 233 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 2700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 2700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 137 MY1= 0 MY2= 741|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 875 AXIAL LOAD= 242|
DESIGNED CAP. KN METER: MOM.= 875 AXIAL CAP.= 243

C O L U M N N O. 1007 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 9180. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION

12 32 MM	3.191	234	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

BRACED /SHORT	in z E.L.z= 2000 mm (3.8.1.3 & 5)
BRACED /SHORT	in y E.L.y= 2000 mm (3.8.1.3 & 5)
END MOMS.	M21= 0 M22= 189 MY1= 0 MY2= 709
SLENDERNESS MOMTS.	KNM: MOMZ= 0 MOMY= 0
DESIGN LOADS KN METER: MOM.	= 894 AXIAL LOAD= 248
DESIGNED CAP. KN METER: MOM.	= 894 AXIAL CAP.= 249

C O L U M N N O. 1008 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2947. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION

40 10 MM	1.039	225	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

BRACED /SHORT	in z E.L.z= 2000 mm (3.8.1.3 & 5)
BRACED /SHORT	in y E.L.y= 2000 mm (3.8.1.3 & 5)
END MOMS.	M21= 0 M22= 133 MY1= 0 MY2= 400
SLENDERNESS MOMTS.	KNM: MOMZ= 0 MOMY= 0
DESIGN LOADS KN METER: MOM.	= 518 AXIAL LOAD= 1210
DESIGNED CAP. KN METER: MOM.	= 518 AXIAL CAP.= 1215

C O L U M N N O. 1009 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 4534. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION

24 16 MM	1.596	224	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

BRACED /SHORT	in z E.L.z= 3650 mm (3.8.1.3 & 5)
BRACED /SHORT	in y E.L.y= 3650 mm (3.8.1.3 & 5)
END MOMS.	M21= 0 M22= 156 MY1= 0 MY2= 490
SLENDERNESS MOMTS.	KNM: MOMZ= 0 MOMY= 0
DESIGN LOADS KN METER: MOM.	= 631 AXIAL LOAD= 1046
DESIGNED CAP. KN METER: MOM.	= 631 AXIAL CAP.= 1050

C O L U M N N O. 1010 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2956. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION

40 10 MM	1.039	230	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

BRACED /SHORT	in z E.L.z= 3650 mm (3.8.1.3 & 5)
BRACED /SHORT	in y E.L.y= 3650 mm (3.8.1.3 & 5)
END MOMS.	M21= 0 M22= 366 MY1= 0 MY2= 142
SLENDERNESS MOMTS.	KNM: MOMZ= 0 MOMY= 0
DESIGN LOADS KN METER: MOM.	= 494 AXIAL LOAD= 1057
DESIGNED CAP. KN METER: MOM.	= 494 AXIAL CAP.= 1063

C O L U M N N O. 1011 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 5084. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
40 16 MM	1.862	230	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

|BRACED /SHORT in z E.L.z= 2700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 2700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 594 MY1= 0 MY2= 149|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 724 AXIAL LOAD= 1297|
|DESIGNED CAP. KN METER: MOM.= 724 AXIAL CAP.= 1320|

C O L U M N N O. 1012 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 6386. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
32 16 MM	2.127	219	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

|BRACED /SHORT in z E.L.z= 2000 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 2000 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 712 MY1= 0 MY2= 180|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 866 AXIAL LOAD= 1488|
|DESIGNED CAP. KN METER: MOM.= 866 AXIAL CAP.= 1491|

C O L U M N N O. 1015 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 4871. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
16 20 MM	1.662	229	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 2700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 2700 mm (3.8.1.3 & 5) |
END MOMS. M21= 0 M22= 512 MY1= 0 MY2= 121 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 623 AXIAL LOAD= 826 |
|DESIGNED CAP. KN METER: MOM.= 623 AXIAL CAP.= 838 |

C O L U M N N O. 1016 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 6219. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
20 20 MM	2.077	219	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 2000 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 2000 mm (3.8.1.3 & 5) |
END MOMS. M21= 0 M22= 586 MY1= 0 MY2= 205 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 772 AXIAL LOAD= 997 |
|DESIGNED CAP. KN METER: MOM.= 772 AXIAL CAP.= 1003 |

C O L U M N N O. 2001 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 4920. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
16 20 MM	2.011	222	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |
END MOMS. M21= 363 M22= 331 MY1= 97 MY2= 100 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 460 AXIAL LOAD= 292 |
|DESIGNED CAP. KN METER: MOM.= 460 AXIAL CAP.= 292 |

C O L U M N N O. 2002 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 9863. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
8 40 MM	4.021	220	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |
END MOMS. M21= 359 M22= 389 MY1= 244 MY2= 526 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 889 AXIAL LOAD= 563 |
|DESIGNED CAP. KN METER: MOM.= 889 AXIAL CAP.= 563 |

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C O L U M N N O. 2003 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 9656. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

20 25 MM 3.927 222 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 288 M22= 361 MY1= 260 MY2= 535|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 873 AXIAL LOAD= 560|
DESIGNED CAP. KN METER: MOM.= 873 AXIAL CAP.= 563

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C O L U M N N O. 2004 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 4174. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

24 16 MM 1.931 220 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 265 M22= 304 MY1= 96 MY2= 100|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 401 AXIAL LOAD= 285|
DESIGNED CAP. KN METER: MOM.= 401 AXIAL CAP.= 288

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C O L U M N N O. 2005 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 7587. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

16 25 MM 3.142 226 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 278 M22= 398 MY1= 220 MY2= 303|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 687 AXIAL LOAD= 407|
DESIGNED CAP. KN METER: MOM.= 687 AXIAL CAP.= 407

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C O L U M N N O. 2008 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 7679. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

16 25 MM 3.142 223 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 268 M22= 428 MY1= 245 MY2= 299|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 710 AXIAL LOAD= 502|
DESIGNED CAP. KN METER: MOM.= 710 AXIAL CAP.= 503

C O L U M N N O. 2009 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5791. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12 25 MM	2.356	224	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 147 M22= 179 MY1= 327 MY2= 360|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 533 AXIAL LOAD= 315|
DESIGNED CAP. KN METER: MOM.= 533 AXIAL CAP.= 320|

C O L U M N N O. 2010 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 9021. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12 32 MM	3.861	223	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 102 M22= 131 MY1= 553 MY2= 721|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 842 AXIAL LOAD= 674|
DESIGNED CAP. KN METER: MOM.= 842 AXIAL CAP.= 678|

C O L U M N N O. 2011 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 9287. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12 32 MM	3.861	224	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 105 M22= 133 MY1= 568 MY2= 740|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 863 AXIAL LOAD= 673|
DESIGNED CAP. KN METER: MOM.= 863 AXIAL CAP.= 675|

C O L U M N N O. 2012 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5651. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12 25 MM	2.356	223	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 160 M22= 181 MY1= 303 MY2= 348|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 522 AXIAL LOAD= 319|
DESIGNED CAP. KN METER: MOM.= 522 AXIAL CAP.= 323|

C O L U M N N O. 2013 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 3738. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
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12	20 MM	1.508	224 EACH END
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(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. MZ1= 113 MZ2= 116 MY1= 215 MY2= 258|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 371 AXIAL LOAD= 306|
|DESIGNED CAP. KN METER: MOM.= 371 AXIAL CAP.= 307|

C O L U M N N O. 2014 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5083. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
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28	16 MM	2.252	219 EACH END
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(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. MZ1= 280 MZ2= 323 MY1= 153 MY2= 109|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 472 AXIAL LOAD= 284|
|DESIGNED CAP. KN METER: MOM.= 472 AXIAL CAP.= 285|

C O L U M N N O. 2015 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5168. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
-------------------	------------	------	----------

28	16 MM	2.252	221 EACH END
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(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. MZ1= 238 MZ2= 308 MY1= 174 MY2= 118|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 477 AXIAL LOAD= 276|
|DESIGNED CAP. KN METER: MOM.= 477 AXIAL CAP.= 276|

C O L U M N N O. 2016 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 3749. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
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12	20 MM	1.508	223 EACH END
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(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. MZ1= 91 MZ2= 114 MY1= 246 MY2= 261|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 371 AXIAL LOAD= 305|
|DESIGNED CAP. KN METER: MOM.= 371 AXIAL CAP.= 305|

*****END OF COLUMN DESIGN RESULTS*****

101J 2975.X 300.X 600 149J

B E A M N O. 1 D E S I G N R E S U L T S - F L E X U R E

LEN - 2975. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1	52.	3- 25 MM	0.	2975.	YES YES
2	528.	2- 50 MM	0.	2975.	YES YES
COMP.		3- 10 MM	(REQD. STEEL=	234. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	632.5/	3382.2	142.46/	575.51	228/	221
248.	704.2/	2541.5	158.18/	488.35	228/	221
496.	766.3/	2009.8	171.29/	405.07	228/	221
744.	816.5/	1553.8	181.80/	325.69	228/	221
992.	854.6/	1154.8	189.70/	250.21	228/	221
1240.	884.4/	805.3	195.84/	179.46	220/	229
1488.	992.5/	582.1	217.89/	131.11	220/	229
1735.	1083.0/	379.0	236.04/	85.37	220/	229
1983.	1155.2/	234.0	250.30/	42.23	220/	229
2231.	1208.3/	234.0	260.66/	1.70	220/	229
2479.	1241.7/	0.0	267.13/	0.00	220/	0
2727.	1257.7/	0.0	270.23/	0.00	219/	0
2975.	1254.7/	0.0	269.64/	0.00	219/	0

B E A M N O. 1 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING	mm	C/C		
mm	kNm	S	T	SIZE	S	T	S+T	S		
END 1	1487	359.4	12	221226	10mm	7	7	15	193	93
1487 END 2	171.0	12	221226	10mm	2	7	11	571	199	135
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 36 mm2 EVENLY DISTRIBUTED										
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *										

B E A M N O. 2 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO SHEAR TORSN LOAD LINK NO. SPACING mm C/C									
mm kN kNm S T SIZE S T S+T S T S+T									
END 1 1762 120.7 12 220226 10mm 3 8 11 571 199 147									
1762 END 2 407.2 12 220226 10mm 13 8 23 130 199 76									
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 369 mm2 EVENLY DISTRIBUTED									
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *									
149J 3525.X 300.X 600 102J									

B E A M N O. 3 D E S I G N R E S U L T S - S H E A R

FROM - TO MAX. SHEAR LOAD LINKS NO. SPACING C/C									
END 1 2500 mm 408.7 kN 221 10 mm 17 156 mm									
2500 4500 mm 178.5 kN 228 10 mm 4 499 mm									
1500 4000 mm 229.5 kN 228 10 mm 5 499 mm									
4000 END 2 396.6 kN 220 10 mm 13 166 mm									

B E A M N O. 3 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	50.	4- 20 MM	0.	6000.	YES	YES
2	528.	2- 50 MM	0.	2900.	YES	NO
3	COMP.	3- 16 MM	(REQD. STEEL= 596. SQ. MM)			
	528.	2- 50 MM	2600.	6000.	NO	YES
COMP.		3- 16 MM	(REQD. STEEL= 598. SQ. MM)			

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	744.7/	3862.8	166.76/	677.41	228/	221
500.	813.7/	2493.9	181.22/	481.31	228/	221
1000.	827.7/	1435.0	184.12/	303.81	228/	221
1500.	822.2/	688.5	182.98/	154.84	220/	229
2000.	873.7/	246.4	193.64/	55.51	220/	229
2500.	804.7/	0.0	179.33/	0.00	220/	0
3000.	769.7/	0.0	172.01/	0.00	221/	0
3500.	1013.2/	234.0	222.07/	3.14	221/	228
4000.	1115.3/	378.5	242.42/	85.26	221/	228
4500.	1092.0/	813.3	237.81/	181.13	221/	228
5000.	1076.2/	1502.5	234.68/	316.32	229/	220
5500.	1079.1/	2537.5	235.25/	487.77	229/	220
6000.	1024.1/	3864.8	224.26/	677.82	229/	220

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO MAX. SHEAR LOAD LINKS NO. SPACING C/C									
END 1 2500 mm 408.7 kN 221 10 mm 17 156 mm									
2500 4500 mm 178.5 kN 228 10 mm 4 499 mm									
1500 4000 mm 229.5 kN 228 10 mm 5 499 mm									
4000 END 2 396.6 kN 220 10 mm 13 166 mm									

102J 6000.X 300.X 600 103J

B E A M N O. S D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING mm	C/C
mm	kN	S T	SIZE	S T	S T	S T
END 1	END 2	0.0	20	0223	10mm	0 15 15
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 579 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

150J	2975.X 300.X 600	151J
15*10c/c199		
3No10 H 45. 0.TO 2975		

3T10	3T10	3T10	3T10	3T10	3T10
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 6 D E S I G N R E S U L T S - F L E X U R E

LEN - 2975. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	52.	3- 25 MM	0.	2975.	YES	YES
2	528.	2- 50 MM	0.	2975.	YES	YES
COMP. 6- 8 MM (REQD. STEEL= 301. SQ. MM)						

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	787.6/ 3567.7	175.76/ 614.85	227/ 222
248.	807.1/ 2868.1	179.84/ 534.54	228/ 221

496.	824.6/ 2322.8	183.47/ 455.30	228/ 221
744.	838.6/ 1847.9	186.39/ 377.74	228/ 221
992.	848.8/ 1425.7	188.50/ 302.08	228/ 221
1240.	853.1/ 1050.5	189.40/ 229.55	228/ 221
1488.	959.2/ 821.6	211.15/ 182.85	220/ 229
1735.	1085.4/ 641.9	236.50/ 144.57	220/ 229
1983.	1192.5/ 479.1	257.58/ 107.92	220/ 229
2231.	1278.0/ 323.9	274.12/ 72.96	220/ 229
2479.	1347.5/ 234.0	287.36/ 39.29	220/ 229
2727.	1394.0/ 234.0	296.13/ 7.30	220/ 229
2975.	1416.7/ 0.0	300.40/ 0.00	220/ 0

B E A M N O. 6 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1487 mm	325.4 kN	222	10 mm	8
1487	END 2	173.2 kN	222	10 mm	5

105J	2975.X 300.X 600	152J
2No50 H 528. 0.TO 2975		
8*10c/c212		
3No25 H 52. 0.TO 2975		

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 7 D E S I G N R E S U L T S - F L E X U R E

LEN - 3525. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	52.	3- 25 MM	0.	3525.	YES	YES
2	528.	2- 50 MM	569.	3525.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1409.0/	0.0	220/ 0
294.	1219.3/	0.0	220/ 0
588.	1004.6/	0.0	220/ 0
881.	845.5/	0.0	222/ 0
1175.	799.7/	0.0	222/ 0
1469.	725.8/	234.0	222/ 227
1763.	625.7/	246.6	222/ 227
2056.	499.6/	437.9	221/ 228
2350.	485.0/	785.7	229/ 220
2644.	486.5/	1213.7	229/ 220
2938.	481.9/	1697.1	229/ 220
3231.	472.1/	2250.4	229/ 220
3525.	459.2/	2893.8	229/ 220

B E A M N O. 7 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1762 mm	112.9 kN	220	10 mm	6	352 mm
1762	END 2	321.3 kN	220	10 mm	11	176 mm

152J	3525.X 300.X 600	106J
2No50	H 528. 569.TO 3525	
6*10c/c352		
3No25	H 52. 0.TO 3525	

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 8 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	48.	4- 16 MM	0.	6000.	YES YES
2	533.	2- 40 MM	0.	2843.	YES NO
3	533.	2- 40 MM	2657.	6000.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	424.0/	2363.9	95.50/	461.64	227/ 222
500.	431.5/	1552.6	97.19/	325.48	227/ 222
1000.	426.2/	885.2	95.99/	196.02	227/ 222
1500.	426.6/	371.7	96.09/	83.72	219/ 230
2000.	554.6/	234.0	124.90/	23.65	219/ 230
2500.	589.5/	0.0	132.77/	0.00	219/ 0
3000.	688.5/	0.0	154.86/	0.00	221/ 0
3500.	803.4/	234.0	179.07/	12.60	222/ 227
4000.	783.1/	281.5	174.82/	63.39	222/ 227
4500.	664.3/	541.6	149.63/	121.99	222/ 227
5000.	641.7/	1023.8	144.53/	224.20	230/ 219
5500.	653.5/	1689.2	147.20/	350.04	230/ 219
6000.	652.5/	2502.5	146.98/	482.59	230/ 219

B E A M N O. 8 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2000 mm	275.1 kN	222	10 mm	8	285 mm
2000	4001 mm	187.5 kN	227	10 mm	4	500 mm
1999	4000 mm	180.2 kN	227	10 mm	4	500 mm
4000	END 2	267.8 kN	219	10 mm	7	333 mm

106J		6000.X 300.X 600		107J
2No40 H 533.	0.TO 2843	2No40 H 533.2657.	TO 6000	
8*10c/c285			7*10c/c333	
4No16 H 148.	0.TO 6000			

OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40
4T16	4T16	4T16	4T16	4T16	4T16
oooo	oooo	oooo	oooo	oooo	oooo

BEAM NO. 9 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	52.	3- 25 MM	0.	6500.	YES YES
2	528.	2- 50 MM	0.	3067.	YES NO
	COMP.	4- 12 MM (REQD. STEEL=		418. SQ. MM)	
3	528.	2- 50 MM	2892.	6500.	NO YES
	COMP.	5- 12 MM (REQD. STEEL=		548. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	963.0/ 3685.0	211.91/ 639.71	227/ 222
542.	896.1/ 2309.7	198.25/ 453.28	227/ 222
1083.	813.0/ 1282.9	181.07/ 275.06	227/ 222
1625.	798.3/ 585.3	178.00/ 131.83	219/ 230
2167.	911.2/ 234.0	201.35/ 42.44	219/ 230
2708.	884.3/ 0.0	195.84/ 0.00	220/ 0
3250.	1036.6/ 0.0	226.77/ 0.00	222/ 0
3792.	1271.5/ 234.0	272.87/ 33.37	222/ 227
4333.	1337.9/ 496.1	285.55/ 111.73	222/ 227
4875.	1252.9/ 908.3	269.29/ 200.76	222/ 227
5417.	1166.8/ 1504.9	252.57/ 316.75	230/ 219
5958.	1257.1/ 2538.3	270.11/ 487.88	230/ 219
6500.	1330.5/ 3814.8	284.14/ 667.23	230/ 219

BEAM NO. 9 DESIGN RESULTS - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2709 mm	347.4 kN	222	10 mm	14	208 mm
2709	5418 mm	174.9 kN	227	10 mm	5	541 mm
1083	4334 mm	224.9 kN	227	10 mm	6	541 mm
4334	END 2	334.3 kN	219	10 mm	10	240 mm

107J	6500.X 300.X 600	108J
=====		
2No50 H 528.	0.TO 3067	2No50 H 528.2892.TO 6500
14*10c/c208		10*10c/c240
3No25 H 52.	0.TO 6500	
=====		

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

BEAM NO. 10 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	56.	2- 32 MM	0.	6500.	YES YES
2	528.	2- 50 MM	0.	3608.	YES NO
	COMP.	3- 10 MM (REQD. STEEL=		234. SQ. MM)	
3	528.	2- 50 MM	3433.	6500.	NO YES
	COMP.	3- 10 MM (REQD. STEEL=		234. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	-------------------------------------	---------------------------------	----------------

0.	1499.6/	3361.5	315.78/	571.13	228/	221
542.	1348.8/	2133.5	287.62/	425.34	228/	221
1083.	1184.5/	1348.4	256.02/	287.54	228/	221
1625.	1149.5/	868.9	249.18/	192.66	220/	229
2167.	1088.2/	505.1	237.06/	113.76	220/	229
2708.	931.7/	234.0	205.54/	42.23	220/	229
3250.	670.3/	0.0	150.97/	0.00	220/	0
3792.	654.8/	0.0	147.47/	0.00	222/	0
4333.	750.4/	337.1	167.95/	75.92	221/	228
4875.	755.6/	710.1	169.05/	159.43	221/	228
5417.	858.5/	1332.7	190.51/	284.57	229/	220
5958.	990.5/	2183.6	217.49/	433.39	229/	220
6500.	1108.3/	3451.4	241.05/	590.20	229/	220

B E A M N O. 10 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1626 mm	272.5 kN	221	10 mm	5	406 mm
1626	3252 mm	217.1 kN	229	10 mm	3	541 mm
3250	4334 mm	210.9 kN	229	10 mm	2	541 mm
4334	END 2	292.9 kN	220	10 mm	8	309 mm
109J		6500.X 300.X 600				110J
2No50 H 528.	0.TO 3608		2No50 H 528.3433.TO 6500			
5*10c/c406						8*10c/c309
2No32 H 56.	0.TO 6500					

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2500 mm	456.3 kN	221	10 mm	19	138 mm
2500	3000 mm	191.7 kN	222	10 mm	2	249 mm
3000	3500 mm	201.7 kN	222	10 mm	2	249 mm
3500	END 2	466.3 kN	220	10 mm	120	131 mm

B E A M N O. 11 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1 52. 3- 25 MM 0. 6000. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing. *

2 533. 2- 40 MM 0. 2811. YES NO

3 453. 2- 40 MM 0. 2811. YES NO

COMP. 4- 20 MM (REQD. STEEL= 1151. SQ. MM)

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing. *

4 533. 2- 40 MM 3189. 6000. NO YES

5 453. 2- 40 MM 3189. 6000. NO YES

COMP. 2- 32 MM (REQD. STEEL= 1479. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF. STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	807.1/ 4043.6	179.83/ 715.72	228/ 221
500.	904.2/ 2585.2	199.93/ 494.74	228/ 221
1000.	946.1/ 1371.8	208.48/ 291.96	228/ 221
1500.	1033.5/ 598.9	226.14/ 134.89	220/ 229
2000.	1170.7/ 234.0	253.33/ 29.85	220/ 229
2500.	1136.4/ 0.0	246.60/ 0.00	220/ 0
3000.	1086.0/ 0.0	236.63/ 0.00	221/ 0
3500.	1291.6/ 0.0	276.72/ 0.00	221/ 0
4000.	1300.6/ 326.3	278.45/ 73.50	221/ 228
4500.	1134.7/ 821.3	246.26/ 182.80	221/ 228
5000.	1035.6/ 1672.7	226.57/ 347.10	229/ 220
5500.	972.1/ 3284.8	213.76/ 554.88	229/ 220
6000.	1084.3/ 4350.9	189.40/ 780.86	229/ 220

B E A M N O. 11 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2500 mm	456.3 kN	221	10 mm	19	138 mm
2500	3000 mm	191.7 kN	222	10 mm	2	249 mm
3000	3500 mm	201.7 kN	222	10 mm	2	249 mm
3500	END 2	466.3 kN	220	10 mm	120	131 mm

110J	6000.X 300.X 600	111J
=====		
2No40 H 533.	10.TO 2811	
19*10c/c138		
3No25 H 152.	10.TO 6000	
=====		

OO		OO						
2T40		2T40		2T40				
3T25		3T25		3T25		3T25		3T25
ooo		ooo		ooo		ooo		ooo

BEAM NO. 12 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1 60. 2- 40 MM 0. 6500. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*

2 528. 2- 50 MM 0. 3067. YES NO

3 428. 1- 50 MM 0. 3067. YES NO

COMP. 2- 40 MM (REQD. STEEL= 2442. SQ. MM)

4 528. 2- 50 MM 2892. 6500. NO YES

5 428. 1- 50 MM 2892. 6500. NO YES

COMP. 2- 40 MM (REQD. STEEL= 2242. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1982.9/ 5249.5	256.30/ 971.34	228/ 221
542.	1235.2/ 3901.6	265.88/ 685.62	228/ 221
1083.	1214.3/ 2110.0	261.83/ 421.53	228/ 221
1625.	1196.2/ 916.8	258.31/ 202.51	220/ 229
2167.	1285.5/ 240.3	275.55/ 54.11	220/ 229
2708.	1184.9/ 0.0	256.10/ 0.00	220/ 0
3250.	1465.7/ 0.0	309.52/ 0.00	221/ 0
3792.	1948.1/ 234.0	394.77/ 24.86	221/ 228
4333.	2223.1/ 628.4	439.68/ 141.54	221/ 228
4875.	2275.2/ 1293.2	447.90/ 277.04	221/ 228
5417.	2141.3/ 2180.5	426.60/ 432.89	229/ 220
5958.	2247.5/ 3835.1	443.54/ 671.52	229/ 220

6500. 2268.6/ 5062.8 446.85/ 931.78 229/ 220

BEAM NO. 12 DESIGN RESULTS - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	1083 mm	543.4 kN	221	10 mm	12	98 mm
1083	3792 mm	216.6 kN	228	10 mm	5	541 mm
3792	END 2	496.4 kN	220	10 mm	23	123 mm

111J	6500.X 300.X 600	112J
=====		
2No50 H	528. 10.TO 3067	23*10c/c123
1No50cHc428.		
2No40 H	160. 10.TO 6500	
=====		

OO		OO						
2T50		2T50		2T50				
2T40		2T40		2T40		2T40		2T40
OO		OO		OO		OO		OO

BEAM NO. 13 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	46.	5- 12 MM	0.	6500.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	0.0/ 0.0	0.00/ 0.00	0/ 0
542.	234.0/ 0.0	32.67/ 0.00	219/ 0
1083.	268.8/ 0.0	60.53/ 0.00	219/ 0

1625.	365.1/	0.0	82.24/	0.00	219/	0
2167.	433.9/	0.0	97.74/	0.00	219/	0
2708.	475.2/	0.0	107.04/	0.00	219/	0
3250.	489.0/	0.0	110.14/	0.00	219/	0
3792.	475.2/	0.0	107.04/	0.00	219/	0
4333.	433.9/	0.0	97.74/	0.00	219/	0
4875.	365.1/	0.0	82.24/	0.00	219/	0
5417.	268.8/	0.0	60.53/	0.00	219/	0
5958.	234.0/	0.0	32.67/	0.00	219/	0
6500.	0.0/	0.0	0.00/	0.00	0/	0

B E A M N O. 13 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX.	SHEAR	LOAD	LINKS	NO.	SPACING	C/C
END 1	1625 mm	62.8	kN	226	10 mm	6	324	mm
4875	END 2	62.8	kN	226	10 mm	6	324	mm

153J 6500.X 300.X 600 154J

6*10c/c324

6*10c/c324

5No12 H 46. 0.TO 6500

5T12	5T12	5T12	5T12	5T12	5T12	5T12
oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 14 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA
					END

1 50. 4- 20 MM 0. 6500. YES YES

2	528.	2- 50 MM	0.	3067.	YES	NO
3	528.	2- 50 MM	3433.	6500.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF	STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)		(KN-METER)	

0.	877.4/	2631.6	194.41/	501.47	227/	222
542.	954.5/	1714.9	210.20/	354.59	227/	222
1083.	971.0/	1034.2	213.55/	226.29	227/	222
1625.	1010.3/	593.1	221.48/	133.59	219/	230
2167.	968.4/	246.5	213.01/	55.52	219/	230
2708.	836.5/	0.0	185.95/	0.00	219/	0
3250.	623.0/	0.0	140.31/	0.00	219/	0
3792.	605.7/	0.0	136.43/	0.00	222/	0
4333.	680.8/	234.0	153.22/	27.54	222/	227
4875.	672.4/	493.9	151.42/	111.25	222/	227
5417.	673.9/	1043.3	151.74/	228.10	230/	219
5958.	633.8/	1783.9	142.75/	366.67	230/	219
6500.	538.7/	2789.9	121.32/	523.82	230/	219

B E A M N O. 14 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX.	SHEAR	LOAD	LINKS	NO.	SPACING	C/C
END 1	1084 mm	288.3	kN	222	10 mm	4	361	mm
1084	1626 mm	219.7	kN	222	10 mm	2	270	mm
1624	4875 mm	204.3	kN	227	10 mm	6	541	mm
4875	END 2	307.3	kN	219	10 mm	7	270	mm

113J 6500.X 300.X 600 114J

2No50 H 528.	0.TO 3067	2No50 H 528.3433.TO 6500	
4*10c/c361			7*10c/c270
4No20 H 50.	0.TO 6500		

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo

=====

B E A M N O. 15 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	4- 20 MM	0.	6000.	YES YES
2	528.	2- 50 MM	0.	2900.	YES NO
	COMP.	3- 10 MM (REQD. STEEL=	234. SQ. MM)		
3	528.	2- 50 MM	3100.	6000.	NO YES
	COMP.	6- 8 MM (REQD. STEEL=	280. SQ. MM)		

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	535.9/ 3465.1	120.70/ 593.09	227/ 222
500.	639.9/ 2037.1	144.13/ 409.60	227/ 222
1000.	694.0/ 1126.7	156.03/ 244.68	227/ 222
1500.	787.1/ 535.4	175.66/ 120.59	219/ 230
2000.	877.3/ 234.0	194.39/ 32.20	219/ 230
2500.	840.3/ 0.0	186.73/ 0.00	219/ 0
3000.	812.3/ 0.0	180.92/ 0.00	222/ 0
3500.	996.2/ 0.0	218.64/ 0.00	222/ 0
4000.	1044.7/ 260.3	228.38/ 58.63	222/ 227
4500.	962.1/ 645.2	211.73/ 145.32	222/ 227
5000.	868.8/ 1236.7	192.64/ 266.16	230/ 219
5500.	819.6/ 2156.2	182.44/ 429.01	230/ 219
6000.	716.1/ 3546.8	160.71/ 610.41	230/ 219

B E A M N O. 15 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
----- ----- ----- ----- ----- ----- -----						
END 1	2000 mm	383.4 kN	222	10 mm	12	181 mm
2000	4500 mm	220.4 kN	227	10 mm	5	499 mm
4499	END 2	379.3 kN	219	10 mm	10	166 mm
----- ----- ----- ----- ----- ----- -----						

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B E A M N O. 16 D E S I G N R E S U L T S - F L E X U R E

LEN - 2225. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

00	00	00		00	00
2T50	2T50	2T50		2T50	2T50
4T20	4T20	4T20		4T20	4T20
oooo	oooo	oooo		oooo	oooo

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1022.3/ 4204.7	223.89/ 749.89	227/ 222
185.	1034.3/ 3839.0	226.31/ 672.36	227/ 222
371.	1038.9/ 3484.2	227.23/ 597.15	227/ 222
556.	1035.8/ 2793.9	226.60/ 524.38	227/ 222
742.	1024.7/ 2315.4	224.39/ 454.16	227/ 222
927.	1005.6/ 1900.0	220.53/ 386.64	227/ 222
1113.	978.1/ 1533.4	214.99/ 321.98	227/ 222
1298.	942.5/ 1205.9	207.74/ 260.19	227/ 222
1483.	898.8/ 910.7	198.81/ 201.25	227/ 222
1669.	870.6/ 665.1	193.02/ 149.80	219/ 230
1854.	859.3/ 470.4	190.68/ 105.94	219/ 230
2040.	836.3/ 282.5	185.90/ 63.63	219/ 230
2225.	802.2/ 234.0	178.81/ 22.82	219/ 230

B E A M N O. 16 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1112 mm	424.1 kN	222	10 mm	9	139 mm
1112	END 2	257.9 kN	222	10 mm	4	370 mm
3No32H 537.	0.TO 2225					
9*10c/c139						4*10c/c370
4No20H 50.	0.TO 2225					

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1112 mm	424.1 kN	222	10 mm	9	139 mm
1112	END 2	257.9 kN	222	10 mm	4	370 mm
3No32H 537.	0.TO 2225					
9*10c/c139						4*10c/c370
4No20H 50.	0.TO 2225					

=====

B E A M N O. 17 D E S I G N R E S U L T S - F L E X U R E

LEN - 2050. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

1	60.	2- 40 MM	0.	2050.	YES YES
2	548.	3- 10 MM	0.	1013.	YES NO
3	547.	5- 12 MM	695.	2050.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	

0.	949.8/	234.0	209.24/	48.99	219/	230
171.	892.0/	234.0	197.40/	14.30	219/	230

342.	825.8/	0.0	183.72/	0.00	219/	0
512.	751.5/	0.0	168.19/	0.00	219/	0
683.	669.6/	0.0	150.81/	0.00	219/	0
854.	761.4/	0.0	170.27/	0.00	222/	0
1025.	908.5/	0.0	200.81/	0.00	222/	0
1196.	1050.2/	0.0	229.49/	0.00	222/	0
1367.	1186.1/	0.0	256.33/	0.00	222/	0
1537.	1315.7/	234.0	281.32/	26.41	222/	227
1708.	1438.5/	241.8	304.46/	54.46	222/	227
1879.	1554.1/	371.8	325.75/	83.75	222/	227
2050.	1662.0/	507.4	345.20/	114.28	222/	227

B E A M N O. 17 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1024 mm	238.3 kN	222	10 mm	6	204 mm
1024	END 2	193.7 kN	219	10 mm	4	341 mm

159J	2050. X 300. X 600	157J
3No10 H 548.	0.TO 10135No12 H 547. 695.TO 2050	
6*10c/c204		4*10c/c341
2No40 H 160.	0.TO 2050	

ooo	ooo	ooo	ooooo	ooooo	ooooo	ooooo
3T10	3T10	3T10	5T12	5T12	5T12	5T12
2T40	2T40	2T40	2T40	2T40	2T40	2T40
oo	oo	oo	oo	oo	oo	oo

=====

B E A M N O. 18 D E S I G N R E S U L T S - F L E X U R E

LEN - 2225. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

1 60. 2- 40 MM 0. 2225. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*

2 533. 2- 40 MM 0. 2225. YES YES

3 453. 2- 40 MM 0. 2225. YES YES

COMP. 4- 20 MM (REQD. STEEL= 1151. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1498.0/ 462.4	315.49/ 104.15	222/ 227
185.	1582.1/ 623.5	330.83/ 140.43	222/ 227
371.	1654.6/ 799.3	343.88/ 178.22	222/ 227
556.	1714.2/ 990.8	354.47/ 217.55	222/ 227
742.	1760.1/ 1197.1	362.52/ 258.47	222/ 227
927.	1790.7/ 1420.1	367.86/ 301.03	222/ 227
1113.	1805.0/ 1662.5	370.34/ 345.28	222/ 227
1298.	1817.9/ 1942.6	372.57/ 393.84	230/ 219
1483.	1866.7/ 2308.6	380.95/ 453.10	230/ 219
1669.	1906.2/ 2727.1	387.69/ 515.06	230/ 219
1854.	1936.7/ 3401.3	392.85/ 579.57	230/ 219
2040.	1958.1/ 3717.2	396.45/ 646.53	230/ 219
2225.	1970.7/ 4043.9	398.56/ 715.79	230/ 219

B E A M N O. 18 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1112 mm	213.4 kN	219	10 mm	4	370 mm
1112	END 2	379.5 kN	219	10 mm	8	158 mm
157J		2225. X 300. X 600				116J
2No40 H 533.	0. TO 2225					
4*10c/c370						8*10c/c158
2No40 H 60.	0. TO 2225					

OO	OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40	2T40
2T40	2T40	2T40	2T40	2T40	2T40	2T40
OO	OO	OO	OO	OO	OO	OO

B E A M N O. 19 D E S I G N R E S U L T S - FLEXURE

LEN - 3525. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	60.	2- 40 MM	0.	3525.	YES YES
2	528.	2- 50 MM	0.	3250.	YES NO
3	547.	5- 12 MM	2389.	3525.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1719.9/ 2918.9	355.47/ 541.38	233/ 224
294.	1617.3/ 2296.8	337.20/ 451.27	233/ 224
587.	1496.8/ 1783.6	315.26/ 366.62	233/ 224
881.	1359.7/ 1347.9	289.67/ 287.44	233/ 224
1175.	1248.0/ 1011.1	268.34/ 221.63	225/ 232
1469.	1117.1/ 716.3	242.79/ 160.75	225/ 232
1762.	962.3/ 459.6	211.78/ 103.51	225/ 232
2056.	785.4/ 234.0	175.31/ 49.94	225/ 232
2350.	592.1/ 234.0	133.37/ 0.03	225/ 232
2644.	381.6/ 0.0	85.96/ 0.00	225/ 0
2937.	505.2/ 0.0	113.80/ 0.00	224/ 0
3231.	662.7/ 234.0	149.26/ 46.74	224/ 233
3525.	804.3/ 467.3	179.25/ 105.25	224/ 233

B E A M N O. 19 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1469 mm	316.1 kN	224	10 mm	6	293 mm
1469	1763 mm	223.0 kN	224	10 mm	2	146 mm
1762	END 2	226.5 kN	225	10 mm	8	251 mm

113J_____3525.X 300.X 600_____153J_____									
=====									
2No50 H 528.	10.T0	3250				15No12 H 547.2389.	T0	3525	
6*10c/c293						8*10c/c251			
2No40 H 60.	10.T0	3525							
=====									
=====									
OO		OO		OO		OO		OO	
2T50		2T50		2T50		2T50		2T50	
2T40		2T40		2T40		2T40		2T40	
OO		OO		OO		OO		OO	
=====									
=====									

B E A M N O. 20 D E S I G N R E S U L T S - F L E X U R E

LEN - 1515. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	4- 20 MM	0.	1515.	YES YES
2	528.	2- 50 MM	0.	1515.	YES YES
COMP. 6- 8 MM (REQD. STEEL= 281. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	797.0/	471.1	177.72/
126.	812.7/	604.1	180.99/
253.	823.3/	744.5	183.21/
379.	862.3/	929.1	191.30/
505.	902.4/	1130.0	199.54/
631.	939.1/	1344.6	207.05/
758.	972.3/	1574.9	213.81/
884.	1002.0/	1823.1	219.82/
1010.	1028.2/	2092.2	225.08/
1136.	1050.8/	2385.7	229.61/
1263.	1069.9/	2708.5	233.42/
1389.	1085.5/	3313.7	236.52/
1515.	1097.6/	3547.7	238.94/

B E A M N O. 20 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	757 mm	289.4 kN	225	10 mm	4	252 mm
757	END 2	397.0 kN	225	10 mm	6	151 mm

153J	1515.X 300.X 600	109J
=====		
2No50 H 528.	0.T0 1515	
4*10c/c252		6*10c/c151
4No20 H 50.	0.T0 1515	
=====		
=====		

OO		OO		OO		OO		OO	
2T50		2T50		2T50		2T50		2T50	
4T20		4T20		4T20		4T20		4T20	
OOOO		OOOO		OOOO		OOOO		OOOO	

B E A M N O. 21 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	52.	3- 25 MM	0.	6650.	YES YES
2	528.	2- 50 MM	0.	3117.	YES NO
COMP. 4- 16 MM (REQD. STEEL= 619. SQ. MM)					

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*
3 533. 2- 40 MM 3623. 6650. NO YES
4 453. 2- 40 MM 3623. 6650. NO YES
COMP. 4- 20 MM (REQD. STEEL= 1156. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	619.3/ 3885.9	97.68/ 682.30	233/ 224
554.	606.6/ 2336.6	136.63/ 457.45	233/ 224
1108.	719.1/ 1182.2	161.34/ 255.57	233/ 224
1662.	901.5/ 484.1	199.37/ 109.03	225/ 232
2217.	1083.6/ 234.0	236.14/ 6.58	225/ 232
2771.	1094.5/ 0.0	238.31/ 0.00	225/ 0
3325.	1169.4/ 0.0	253.07/ 0.00	224/ 0
3879.	1359.6/ 0.0	289.66/ 0.00	224/ 0
4433.	1360.9/ 234.0	289.90/ 52.52	224/ 233
4988.	1182.0/ 682.0	255.53/ 153.46	224/ 233
5542.	984.7/ 1387.4	216.31/ 294.90	232/ 225
6096.	871.1/ 2582.6	193.10/ 494.36	232/ 225
6650.	782.1/ 4048.7	155.66/ 716.81	232/ 225

B E A M N O. 21 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2771 mm	424.2 kN	224	10 mm	20	145 mm
2771	4434 mm	166.3 kN	233	10 mm	3	554 mm
2217	4434 mm	237.4 kN	233	10 mm	4	554 mm
4434	END 2	419.8 kN	225	10 mm	15	158 mm
109J		6650.X 300.X 600				105J
2No50 H 528.	10.TO 3117			2No40 H 533.3623	TO 6650	
20*10c/c145						15*10c/c158
3No25 H 152.	10.TO 6650					
OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T40	2T40	2T40	2T40
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

105J		2885.X 300.X 600		150J	
=====					
3No32 H 537.	0.TO 2885				
16*10c/c 85					10*10c/c144
3No25 H 52.	0.TO 2885				
=====					

OOO	OOO	OOO	OOO	OOO	OOO
3T32	3T32	3T32	3T32	3T32	3T32
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

BEAM NO. 23 DESIGN RESULTS - FLEXURE

LEN - 2425. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	60.	2- 40 MM	0.	2425.	YES YES
2	528.	2- 50 MM	0.	2425.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	551.1/	0.0	224/ 0
202.	692.6/	0.0	224/ 0
404.	830.7/	234.0	224/ 233
606.	959.3/	292.2	224/ 233
808.	1077.9/	478.8	224/ 233
1010.	1185.8/	673.2	224/ 233
1212.	1282.3/	890.4	224/ 233
1415.	1366.9/	1124.8	224/ 233
1617.	1438.8/	1378.6	224/ 233
1819.	1527.2/	1685.3	232/ 225
2021.	1624.1/	2045.6	232/ 225
2223.	1713.0/	2453.3	232/ 225
2425.	1793.6/	2923.9	232/ 225

BEAM NO. 23 DESIGN RESULTS - SHEAR

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO mm	SHEAR TORSN kNm	LOAD S T	LINK SIZE	NO. S T	SPACING mm C/C
END 1 1212	183.5	9	225225	10mm	2 6 8
1212 END 2	337.1	9	225222	10mm	5 6 12

1212 END 2	337.1	9	225222	10mm	5 6 12
1212 END 2	337.1	9	225222	10mm	5 6 12

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 581 mm2 EVENLY DISTRIBUTED
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *

150J 2425.X 300.X 600 101J

2No50/H 528.	0.TO 2425				
8*10c/c 135					
2No40/H 60.	0.TO 2425				

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
2T40	2T40	2T40	2T40	2T40	2T40
OO	OO	OO	OO	OO	OO

BEAM NO. 24 DESIGN RESULTS - FLEXURE

LEN - 2885. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	3- 20 MM	0.	2885.	YES YES
---	-----	----------	----	-------	---------

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	0.0/	0.0	0/ 0
240.	234.0/	0.0	226/ 0
481.	263.7/	0.0	226/ 0
721.	387.0/	0.0	226/ 0

962.	501.6/	0.0	112.98/	0.00	226/	0
1202.	604.7/	0.0	136.20/	0.00	226/	0
1442.	694.8/	0.0	156.18/	0.00	226/	0
1683.	772.3/	0.0	172.55/	0.00	226/	0
1923.	832.3/	0.0	185.07/	0.00	226/	0
2164.	876.1/	0.0	194.15/	0.00	226/	0
2404.	904.7/	0.0	200.02/	0.00	226/	0
2645.	918.7/	0.0	202.90/	0.00	226/	0
2885.	919.7/	0.0	203.09/	0.00	226/	0

B E A M N O. 24 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2163 mm	126.4 kN	226	10 mm	7	360 mm
2163	END 2	10.8 kN	223	10 mm	3	360 mm
Nominal links should be provided through out the length.						

152J	2885.X 300.X 600	151J
7*10c/c360		3*10c/c360
3No20 H 50.	0.TO 2885	

3T20	3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 25 D E S I G N R E S U L T S - F L E X U R E

LEN - 2425. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
-------	--------------	----------	------------	----------	-------------------

1 50. 3- 20 MM 0. 2425. YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	930.0/	0.0	205.19/ 0.00 223/ 0
202.	884.6/	0.0	195.89/ 0.00 223/ 0
404.	831.4/	0.0	184.89/ 0.00 223/ 0
606.	770.5/	0.0	172.19/ 0.00 223/ 0
808.	702.8/	0.0	157.88/ 0.00 223/ 0
1010.	630.2/	0.0	141.95/ 0.00 223/ 0
1212.	553.2/	0.0	124.59/ 0.00 223/ 0
1415.	470.3/	0.0	105.92/ 0.00 223/ 0
1617.	382.7/	0.0	86.19/ 0.00 223/ 0
1819.	291.1/	0.0	65.56/ 0.00 223/ 0
2021.	234.0/	0.0	44.21/ 0.00 223/ 0
2223.	234.0/	0.0	22.31/ 0.00 223/ 0
2425.	0.0/	0.0	0.00/ 0.00 0/ 0

B E A M N O. 25 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1212 mm	41.8 kN	223	10 mm	5	303 mm
1212	END 2	111.1 kN	223	10 mm	5	303 mm

151J	2425.X 300.X 600	149J
5*10c/c303		5*10c/c303
3No20 H 50.	0.TO 2425	

3T20	3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo	ooo

=====

B E A M N O. 26 D E S I G N R E S U L T S - F L E X U R E

LEN - 3525. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	60.	2- 40 MM	0.	3525.	YES YES
2	528.	2- 50 MM	0.	2956.	YES NO
	COMP.	3- 12 MM	(REQD. STEEL=	318. SQ. MM)	
3	548.	5- 10 MM	2389.	3525.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	1711.7/	3584.9	354.02/	618.50	234/ 223
294.	1631.7/	2695.9	339.78/	510.65	234/ 223
587.	1533.5/	2026.2	322.00/	407.80	234/ 223
881.	1417.3/	1470.7	300.51/	310.44	234/ 223
1175.	1353.5/	1065.0	288.50/	232.43	225/ 232
1469.	1271.8/	719.5	272.92/	161.44	225/ 232
1762.	1155.4/	420.3	250.32/	94.68	225/ 232
2056.	1003.9/	234.0	220.20/	32.33	225/ 232
2350.	816.5/	0.0	181.80/	0.00	225/ 0
2644.	596.4/	0.0	134.33/	0.00	225/ 0
2937.	777.8/	0.0	173.70/	0.00	223/ 0
3231.	955.0/	234.0	210.30/	25.74	223/ 234
3525.	1099.1/	387.0	239.21/	87.17	223/ 234

B E A M N O. 26 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO. SPACING C/C
----- ----- ----- ----- ----- -----					
END 1	1762 mm	375.0 kN	223	10 mm	11 176 mm
1762	END 2	256.2 kN	226	10 mm	10 195 mm
----- ----- ----- ----- ----- -----					

_____ 114J _____ 3525. X 300. X 600 _____ 154J _____

=====											
2No50 H	528.	0.	TO	2956						5No10 H	548.2389. TO 3525
11*10c/c	176									10*10c/c	195
2No40 H	160.	0.	TO	3525							
=====											

OO		OO		OO		OO		OO		OO	
2T50		2T50		2T50		2T50		2T50		2T50	
2T40		2T40		2T40		2T40		2T40		2T40	
OO		OO		OO		OO		OO		OO	

=====

B E A M N O. 27 D E S I G N R E S U L T S - F L E X U R E

LEN - 1515. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	52.	3- 25 MM	0.	1515.	YES YES
2	528.	2- 50 MM	0.	1515.	YES YES
	COMP.	5- 10 MM	(REQD. STEEL=	378. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	1106.4/	383.0	240.68/	86.26	223/ 234
126.	1118.2/	522.3	243.01/	117.65	223/ 234
253.	1123.1/	665.3	243.96/	149.85	223/ 234
379.	1120.9/	821.6	243.55/	182.85	223/ 234
505.	1167.8/	1040.1	252.77/	227.48	231/ 226
631.	1212.2/	1275.7	261.41/	273.69	231/ 226
758.	1252.6/	1529.5	269.24/	321.26	231/ 226
884.	1289.5/	1803.4	276.32/	370.07	231/ 226
1010.	1322.5/	2101.4	282.62/	420.14	231/ 226
1136.	1351.8/	2427.9	288.18/	471.40	231/ 226
1263.	1377.6/	2788.9	293.05/	523.69	231/ 226
1389.	1399.8/	3389.1	297.24/	576.99	231/ 226
1515.	1418.7/	3644.8	300.77/	631.20	231/ 226

B E A M N O. 27 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	757 mm	319.0 kN	226	10 mm	5	189 mm
757	END 2	432.7 kN	226	10 mm	7	126 mm
154J			1515.X 300.X 600			110J
2No50 H 528.	0.TO 1515					
5*10c/c189						7*10c/c126
3No25 H 52.	0.TO 1515					
OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 28 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
1	50.	4- 20 MM	0.	6650.	YES	YES
2	528.	2- 50 MM	0.	3117.	YES	NO
COMP.	4- 12 MM	(REQD. STEEL=	429. SQ. MM)			
3	528.	2- 50 MM	3533.	6650.	NO	YES
COMP.	3- 10 MM	(REQD. STEEL=	234. SQ. MM)			

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
----------------	---------------------------------	------------------------------	----------------

0.	605.5/	3695.9	136.39/	642.03	234/	223
554.	632.3/	2331.8	142.42/	456.70	234/	223
1108.	642.1/	1310.7	144.63/	280.39	234/	223
1662.	743.5/	643.5	166.50/	144.94	226/	231
2217.	955.7/	277.8	210.44/	62.57	226/	231
2771.	1025.6/	0.0	224.55/	0.00	226/	0
3325.	934.9/	0.0	206.21/	0.00	204/	0
3879.	1067.1/	0.0	232.87/	0.00	224/	0
4433.	1047.5/	234.0	228.95/	46.27	223/	234
4988.	881.5/	539.4	195.26/	121.48	223/	234
5542.	766.2/	1090.0	171.28/	237.41	231/	226
6096.	789.8/	2000.2	176.23/	403.47	231/	226
6650.	795.2/	3396.5	177.35/	578.56	231/	226

B E A M N O. 28 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING mm C/C
mm	kN	kNm	S	T	S	T
END 1	2771	337.7	11	223226	10mm	12 14 27 230 199 107
2771	4988	169.3	11	234226	10mm	4 12 15 554 199 146
4987	END 2	319.2	11	234220	10mm	6 8 15 243 199 104
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 350 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

110J 6650.X 300.X 600 106J

2No50 H 528.	10.TO 3117					2No50 H 528.3533.TO 16650
27*10c/c107						15*10c/c104
4No20 H 150.	10.TO 6650					

OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50
4T20	4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo	oooo

=====	
106J	5310.X 300.X 600
=====	
2No50 H 528.	10.TO 5310
23*10c/c105	123*10c/c116
2No40 H 160.	10.TO 5310
=====	
=====	

B E A M N O. 29 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	60.	2- 40 MM	0.	5310.	YES YES
2	528.	2- 50 MM	0.	5310.	YES YES
COMP. 5- 12 MM (REQD. STEEL= 537. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1741.0/ 3663.6	359.18/ 635.16	233/ 224
442.	1441.9/ 2522.8	305.10/ 485.59	233/ 224
885.	1148.5/ 1636.4	248.97/ 340.62	233/ 224
1328.	853.6/ 924.3	189.50/ 204.03	233/ 224
1770.	647.9/ 440.0	145.93/ 99.09	226/ 231
2212.	437.5/ 234.0	98.54/ 11.05	226/ 231
2655.	551.2/ 234.0	124.14/ 19.76	224/ 233
3098.	901.6/ 453.8	199.38/ 102.20	224/ 233
3540.	1191.0/ 861.3	257.29/ 191.08	224/ 233
3982.	1415.3/ 1338.2	300.12/ 285.60	224/ 233
4425.	1621.0/ 1924.5	337.87/ 390.79	232/ 225
4868.	1940.2/ 2802.1	393.44/ 525.51	232/ 225
5310.	2269.3/ 3803.6	446.97/ 664.86	232/ 225

B E A M N O. 29 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO mm	SHEAR kNm	TORSN kNm	LOAD kN	LINK SIZE	NO. T S+T	SPACING mm C/C

END 1 2213	340.5	9	224233	10mm	10 12 23	221 199 105
2213 2656	225.1	9	224233	10mm	2 3 4	221 199 105
2656 3097	202.0	9	224229	10mm	2 3 4	221 199 105
3097 END 2	317.4	9	225229	10mm	8 12 23	276 199 116

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 457 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

B E A M N O. 30 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1137 mm	459.1 kN	223	10 mm	10	126 mm
1137	END 2	262.0 kN	223	10 mm	4	379 mm
115J 2275.X 300.X 600 155J						
2No50 H 528.	0.TO 2275					
10*10c/c126						4*10c/c379
3No25 H 52.	0.TO 2275					
OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 31 D E S I G N R E S U L T S - F L E X U R E

LEN - 2765. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	52.	3- 25 MM	0.	2765.	YES	YES
2	528.	2- 50 MM	22.	2765.	NO	YES
COMP. 3- 16 MM (REQD. STEEL= 587. SQ. MM)						

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1309.7/	0.0	280.18/ 0.00
230.	1055.7/	0.0	230.58/ 0.00

461.	939.2/	0.0	207.09/	0.00	223/	0
691.	1013.5/	0.0	222.14/	0.00	223/	0
922.	1057.7/	234.0	231.00/	11.72	223/	234
1152.	1077.5/	292.4	234.93/	65.85	224/	233
1383.	1069.5/	546.6	233.33/	123.11	224/	233
1613.	1049.6/	841.2	229.37/	186.93	232/	225
1843.	1115.8/	1289.0	242.54/	276.23	232/	225
2074.	1169.0/	1803.9	253.00/	370.15	232/	225
2304.	1209.6/	2407.1	260.91/	468.24	232/	225
2535.	1237.7/	3357.2	266.35/	570.22	232/	225
2765.	1254.4/	3853.4	269.58/	675.41	232/	225

B E A M N O. 31 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1382 mm	203.6 kN	225	10 mm	5	345 mm
1382	END 2	462.7 kN	225	10 mm	14	106 mm
155J 2765.X 300.X 600 111J						
2No50 H 528.	22.TO 2765					
5*10c/c345						14*10c/c106
3No25 H 52.	0.TO 2765					
OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 32 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	50.	4- 20 MM	0.	6650.	YES	YES
2	528.	2- 50 MM	0.	3117.	YES	NO
	COMP.	4- 12 MM	(REQD. STEEL=	423. SQ. MM)		
3	528.	2- 50 MM	3533.	6650.	NO	YES
	COMP.	3- 10 MM	(REQD. STEEL=	234. SQ. MM)		

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	571.4/ 3689.3	128.69/ 640.63	233/ 224
554.	604.2/ 2331.2	136.08/ 456.61	233/ 224
1108.	620.0/ 1317.2	139.64/ 281.61	233/ 224
1662.	701.8/ 631.7	157.67/ 142.29	225/ 232
2217.	921.4/ 273.9	203.44/ 61.70	225/ 232
2771.	999.8/ 0.0	219.37/ 0.00	225/ 0
3325.	908.1/ 0.0	200.72/ 0.00	204/ 0
3879.	1027.9/ 0.0	225.02/ 0.00	224/ 0
4433.	1002.2/ 234.0	219.85/ 43.13	224/ 233
4988.	831.2/ 519.4	184.84/ 116.99	224/ 233
5542.	719.6/ 1070.1	161.44/ 233.46	232/ 225
6096.	734.5/ 1965.5	164.61/ 397.69	232/ 225
6650.	731.4/ 3360.6	163.95/ 570.95	232/ 225

B E A M N O. 32 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2217 mm	335.3 kN	224	10 mm	11	221 mm
2217	4434 mm	229.1 kN	233	10 mm	4	554 mm
4434	END 2	209.6 kN	233	10 mm	4	554 mm
4434	END 2	315.9 kN	225	10 mm	9	277 mm

111J	6650.X 300.X 600	107J
2No50 H 528.	0.TO 3117	2No50 H 528.3533.TO 6650
11*10c/c221		9*10c/c277
4No20 H 50.	0.TO 6650	

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 33 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	60.	2- 40 MM	0.	5310.	YES YES
2	528.	2- 50 MM	0.	5310.	YES YES
	COMP.	4- 12 MM	(REQD. STEEL=	418. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1593.1/ 3528.8	332.83/ 606.61	233/ 224
442.	1335.7/ 2356.1	285.13/ 460.45	233/ 224
885.	1079.8/ 1516.5	235.39/ 318.87	233/ 224
1328.	818.6/ 836.1	182.24/ 185.87	233/ 224
1770.	670.4/ 399.9	150.99/ 90.06	225/ 232
2212.	492.8/ 234.0	110.98/ 6.82	225/ 232
2655.	577.8/ 234.0	130.14/ 9.40	223/ 234
3098.	890.0/ 386.4	197.01/ 87.02	223/ 234
3540.	1125.3/ 768.8	244.40/ 171.83	223/ 234
3982.	1289.3/ 1217.4	276.28/ 262.43	223/ 234
4425.	1451.7/ 1786.4	306.92/ 367.11	231/ 226
4868.	1716.6/ 2628.9	354.88/ 501.08	231/ 226
5310.	1984.3/ 3684.8	400.83/ 639.66	232/ 225

B E A M N O. 33 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2213 mm	332.7 kN	224	10 mm	10	245 mm
2213	2656 mm	210.1 kN	232	10 mm	2	221 mm
2654	3097 mm	193.2 kN	232	10 mm	2	221 mm
3097	END 2	315.8 kN	225	10 mm	9	276 mm

B E A M N O. 38 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO SHEAR TORSN LOAD LINK NO. SPACING mm C/C									
mm	kNm	S	T	SIZE	S	T	S+T	S	T
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
END 1	2217	403.2	17	223226	10mm	14	12	27	158 199 88
2217	4434	220.8	17	234226	10mm	4	12	15	554 199 146
2217	4434	215.3	17	234221	10mm	4	12	15	554 199 146
4434	END 2	397.7	17	226221	10mm	13	12	27	170 199 92
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 361 mm2 EVENLY DISTRIBUTED									
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

112J 6650.X 300.X 600 108J									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
2No50 H 528. 10.TO 3117 2No50 H 528.3533.TO 6650									
27*10c/c 88 127*10c/c 92									
4No20 H 50. 10.TO 6650									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

OO	OO	OO	OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50
4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 39 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
----- ----- ----- ----- ----- ----- -----						
1	52.	3- 25 MM	0.	5310.	YES	YES
2	528.	2- 50 MM	0.	2670.	YES	NO
(REQD. STEEL= 370. SQ. MM)						
3	528.	2- 50 MM	2198.	5310.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	842.5/ 3636.5	187.20/ 629.43	234/ 223

442.	853.3/ 2350.4	189.43/ 459.57	234/ 223
885.	821.2/ 1435.2	182.77/ 303.85	234/ 223
1328.	743.6/ 732.8	166.51/ 164.25	234/ 223
1770.	752.0/ 311.1	168.30/ 70.08	226/ 231
2212.	671.1/ 0.0	151.15/ 0.00	226/ 0
2655.	628.9/ 0.0	141.64/ 0.00	223/ 0
3098.	908.7/ 234.0	200.85/ 5.40	223/ 234
3540.	1095.3/ 341.6	238.46/ 76.94	223/ 234
3982.	1186.8/ 709.3	256.48/ 159.27	223/ 234
4425.	1190.2/ 1161.9	257.13/ 251.61	223/ 234
4868.	1280.5/ 1899.2	274.61/ 386.49	231/ 226
5310.	1327.4/ 2878.1	283.56/ 535.89	231/ 226

B E A M N O. 39 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO MAX. SHEAR LOAD LINKS NO. SPACING C/C									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
END 1 884 mm 398.4 kN 223 10 mm 7 147 mm									
884 3540 mm 212.0 kN 234 10 mm 5 531 mm									
3540 END 2 352.2 kN 226 10 mm 9 221 mm									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
108J 5310.X 300.X 600 104J									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
2No50 H 528. 0.TO 2670 2No50 H 528.2198.TO 5310									
7*10c/c147 9*10c/c221									
3No25 H 52. 0.TO 5310									
===== ===== ===== ===== ===== ===== ===== ===== ===== =====									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

OO	OO	OO	OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo
1	1	1	1	1	1	1	1	1	1

155J 2225.X 300.X 600 160J

B E A M N O. 40 D E S I G N R E S U L T S - F L E X U R E

LEN - 2225. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1 50. 4- 20 MM 0. 2225. YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
-------------------	-----------------------------------	--------------------------------	---------------

0.	0.0/	0.0	0.00/	0.00	0/	0
185.	234.0/	0.0	25.81/	0.00	219/	0
371.	234.0/	0.0	51.17/	0.00	219/	0
556.	336.5/	0.0	75.80/	0.00	219/	0
742.	441.7/	0.0	99.48/	0.00	219/	0
927.	541.3/	0.0	121.93/	0.00	219/	0
1113.	634.3/	0.0	142.86/	0.00	219/	0
1298.	722.8/	0.0	162.14/	0.00	219/	0
1483.	805.5/	0.0	179.50/	0.00	219/	0
1669.	881.6/	0.0	195.27/	0.00	219/	0
1854.	951.1/	0.0	209.51/	0.00	219/	0
2040.	1014.4/	0.0	222.31/	0.00	219/	0
2225.	1071.9/	0.0	233.82/	0.00	219/	0

B E A M N O. 40 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	1112 mm	139.9 kN	219	10 mm	4	370 mm
1112	END 2	58.8 kN	219	10 mm	4	370 mm

B E A M N O. 41 D E S I G N R E S U L T S - F L E X U R E

LEN - 2050. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1 50. 4- 20 MM 0. 2050. YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
-------------------	-----------------------------------	--------------------------------	---------------

0.	1033.7/	0.0	226.19/	0.00	222/	0
171.	1054.5/	0.0	230.35/	0.00	222/	0
342.	1070.4/	0.0	233.51/	0.00	222/	0
512.	1081.3/	0.0	235.69/	0.00	222/	0
683.	1087.3/	0.0	236.89/	0.00	222/	0
854.	1088.3/	0.0	237.09/	0.00	222/	0
1025.	1084.4/	0.0	236.30/	0.00	222/	0
1196.	1075.5/	0.0	234.53/	0.00	222/	0
1367.	1061.6/	0.0	231.77/	0.00	222/	0
1537.	1042.9/	0.0	228.02/	0.00	222/	0
1708.	1019.3/	0.0	223.30/	0.00	221/	0
1879.	991.1/	0.0	217.61/	0.00	221/	0
2050.	966.1/	0.0	212.54/	0.00	219/	0

1083.	335.1/	1967.6	75.49/	398.04	228/	221
1625.	496.7/	1353.8	111.88/	288.56	228/	221
2167.	629.5/	847.9	141.80/	188.31	228/	221
2708.	753.1/	474.7	168.53/	106.92	220/	229
3250.	881.7/	234.0	195.30/	50.78	220/	229
3792.	939.4/	234.0	207.11/	5.87	219/	230
4333.	932.7/	0.0	205.76/	0.00	219/	0
4875.	863.0/	0.0	191.43/	0.00	219/	0
5417.	750.0/	0.0	167.88/	0.00	219/	0
5958.	648.5/	0.0	146.07/	0.00	223/	0
6500.	625.1/	0.0	140.80/	0.00	221/	0

B E A M N O. 104 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING	mm	C/C
mm	mm	kN	kNm	S	T	S	T	S	T
END 1	2709	225.7	20	221225	10mm	8	14	23	338
2709	5418	145.2	20	221225	10mm	5	14	18	541
5416	END 2	71.1	20	221225	10mm	1	5	8	571
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 578 mm2 EVENLY DISTRIBUTED									
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *									

122J 6500.X 300.X 600 123J

2No50 H 528.	10.TO 4692								
23*10c/c125									8*10c/c135
3No20 H 150.	10.TO 6500								

OO	OO	OO	OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 105 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR	INFO	FROM	TO	ANCHOR
	mm			mm	mm	STA
						END

1	50.	4- 20 MM	0.	6000.	YES	YES
---	-----	----------	----	-------	-----	-----

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	
0.	351.7/	0.0	79.21/
500.	554.6/	0.0	124.90/
1000.	745.8/	0.0	166.98/
1500.	921.7/	0.0	203.51/
2000.	1063.5/	0.0	232.14/
2500.	1160.0/	0.0	251.24/
3000.	1194.2/	0.0	257.91/
3500.	1171.5/	0.0	253.49/
4000.	1084.9/	0.0	236.40/
4500.	952.5/	0.0	209.79/
5000.	785.2/	0.0	175.27/
5500.	600.3/	0.0	135.20/
6000.	406.3/	0.0	91.52/

B E A M N O. 105 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C
END 1	2499 mm	98.4 kN	221	10 mm	8	357	mm
3499	END 2	95.8 kN	220	10 mm	8	357	mm

123J 6000.X 300.X 600 124J

8*10c/c357									8*10c/c357
4No20 H 50.	0.TO 6000								

4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O . 106 D E S I G N R E S U L T S - F L E X U R E						
LEN -	6500. mm	FY - 460.	FC - 40.	SIZE -	300. X	600. mm
LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR	STA END
1	50.	4- 20 MM	0.	6500.	YES	YES
2	528.	2- 50 MM	1808.	6500.	NO	YES
	COMP.	3- 12 MM	(REQD. STEEL=	332. SQ. MM)		

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	663.7/	149.48/	223/ 0
542.	692.6/	155.73/	224/ 0
1083.	782.5/	174.70/	221/ 0
1625.	890.7/	197.15/	221/ 0
2167.	955.4/	210.37/	221/ 0
2708.	956.6/	210.61/	221/ 228
3250.	886.0/	196.18/	222/ 227
3792.	752.5/	168.40/	222/ 227
4333.	614.4/	138.39/	230/ 219
4875.	477.9/	107.64/	230/ 219
5417.	312.6/	70.42/	230/ 219
5958.	234.0/	28.11/	230/ 219
6500.	332.1/	0.00/	0/ 219

BEAM NO. 106 DESIGN RESULTS - SHEAR

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING
mm	kN	kNm	S	T	SIZE	S T S+T
END 1	1093	69.2	21	222226	10mm	1 5 8 571 199 135
1083	4334	171.8	21	1219226	10mm	6 17 22 541 199 146
4334	END 2	224.4	21	1219226	10mm	6 11 8 361 199 128

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 281 mm2 EVENLY DISTRIBUTED

* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *

B E A M N O. 107 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING mm C/C
mm	kN	mm	S T SIZE	S T S+T	S T S+T
END 1 1625	175.2	21	222225	10mm	2 8 10 571 199 148
1625 END 2	194.6	21	222225	10mm	8 24 32 571 199 148
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 581 mm2 EVENLY DISTRIBUTED					
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *					

126J	6500.X 300.X 600	127J
2No40 H 533.	10.TO 3009	12No40 H 533.3491.TO 16500
10*10c/c148		11*32*10c/c148
4No20 H 150.	10.TO 6500	

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
mm	kN	mm	mm	mm	mm
END 1 3999	157.3	222	10	11	399
1999 END 2	154.2	219	10	11	399

B E A M N O. 108 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
1	46.	5- 12 MM	0.	6000.	YES	YES
2	537.	2- 32 MM	0.	2843.	YES	NO
3	541.	3- 25 MM	3157.	6000.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	467.0/ 1485.9	105.18/ 313.25	227/ 222
500.	468.7/ 1080.9	105.57/ 235.61	227/ 222

1000.	459.1/	720.1	103.40/	161.56	227/ 222
1500.	439.3/	421.7	98.95/	94.97	219/ 230
2000.	427.0/	234.0	96.17/	42.88	219/ 230
2500.	369.7/	0.0	83.28/	0.00	219/ 0
3000.	259.4/	0.0	58.43/	0.00	219/ 0
3500.	370.2/	0.0	83.39/	0.00	222/ 0
4000.	434.2/	234.0	97.79/	39.23	222/ 227
4500.	453.2/	400.5	102.09/	90.22	222/ 227
5000.	472.7/	684.9	106.46/	154.09	230/ 219
5500.	487.3/	1035.9	109.75/	226.62	230/ 219
6000.	490.5/	1429.3	110.47/	302.75	230/ 219

B E A M N O. 108 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
mm	kN	mm	mm	mm	mm
END 1 3999	157.3	222	10	11	399
1999 END 2	154.2	219	10	11	399

127J	6000.X 300.X 600	128J
2No32 H 537.	0.TO 2843	3No25 H 541.3157.TO 6000
11*10c/c399		11*10c/c399
5No12 H 46.	0.TO 6000	

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
mm	kN	mm	mm	mm	mm
END 1 3999	157.3	222	10	11	399
1999 END 2	154.2	219	10	11	399

B E A M N O. 109 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
0.	467.0/ 1485.9	105.18/ 313.25	227/ 222			
500.	468.7/ 1080.9	105.57/ 235.61	227/ 222			

1	50.	3- 20 MM	0.	6500.	YES	YES
2	533.	2- 40 MM	0.	3009.	YES	NO
3	537.	2- 32 MM	3491.	6500.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	453.4/ 1718.0	102.11/ 355.13	228/ 221
542.	477.1/ 1186.9	107.46/ 256.48	228/ 221
1083.	487.2/ 109.74/ 722.8	162.13/ 722.8	228/ 221
1625.	499.0/ 357.9	112.38/ 80.62	220/ 229
2167.	508.0/ 234.0	114.42/ 14.51	220/ 229
2708.	461.4/ 0.0	103.93/ 0.00	220/ 0
3250.	524.3/ 0.0	118.09/ 0.00	222/ 0
3792.	681.2/ 0.0	153.29/ 0.00	222/ 0
4333.	777.6/ 234.0	173.67/ 38.30	222/ 227
4875.	815.2/ 428.7	181.53/ 96.56	222/ 227
5417.	811.8/ 723.4	180.81/ 162.27	230/ 219
5958.	838.8/ 1137.4	186.43/ 246.79	230/ 219
6500.	851.1/ 1608.5	188.98/ 335.60	230/ 219

B E A M N O. 109 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING	mm C/C
mm	kN	kNm	S T	SIZE	S T	S T	S T
END 1	4333	184.4	22	222226	10mm	7 21 29	571 199 145
2166 END 2	166.2	22	222226	10mm	7 21 29	571 199 145	
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 580 mm2 EVENLY DISTRIBUTED							
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *							

128J	6500.X 300.X 600	129J
2No40 H 533.	10.TO 3009	2No32 H 537.3491.TO 6500
29*10c/c145		129*10c/c145
3No20 H 50.	10.TO 6500	
OO	OO	OO
2T40	2T40	2T32
3T20	3T20	3T20
ooo	ooo	ooo

B E A M N O. 110 D E S I G N R E S U L T S - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	48.	4- 16 MM	0.	6500.	YES YES
2	541.	3- 25 MM	0.	3551.	YES NO
3	543.	4- 20 MM	3491.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	796.3/ 1259.8	177.58/ 270.62	227/ 222
542.	742.6/ 942.4	166.31/ 207.73	227/ 222
1083.	679.2/ 656.3	152.88/ 147.82	227/ 222
1625.	635.2/ 440.4	143.06/ 99.19	219/ 230
2167.	567.6/ 243.3	127.84/ 54.80	219/ 230
2708.	468.7/ 234.0	105.58/ 15.38	219/ 230
3250.	335.1/ 0.0	75.47/ 0.00	219/ 0
3792.	275.4/ 0.0	62.03/ 0.00	222/ 0
4333.	360.4/ 234.0	81.17/ 19.95	222/ 227
4875.	414.1/ 298.0	93.26/ 67.12	222/ 227
5417.	449.9/ 532.4	101.33/ 119.91	230/ 219
5958.	497.1/ 822.0	111.97/ 182.95	230/ 219
6500.	534.8/ 1148.4	120.46/ 248.96	230/ 219

B E A M N O. 110 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING	mm C/C
mm	kN	kNm	S T	SIZE	S T	S T	S T
END 1	1625	118.0	9	222223	10mm	2 8 10	571 199 148
1625 END 2	123.8	9	222223	10mm	8 24 32	571 199 148	
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 473 mm2 EVENLY DISTRIBUTED							
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *							

130J	6500.X 300.X 600		131J
=====			
3No25 H 541.	0.TO 3551	4No20 H 543.3491.	TO 6500
=====			
10*10c/c148			32*10c/c148
=====			
=====			
4No16 H 48.	10.TO 6500		
=====			

ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo
3T25	3T25	3T25	3T25	3T25	4T20	4T20	4T20	4T20	4T20
4T16	4T16	4T16	4T16	4T16	4T16	4T16	4T16	4T16	4T16
oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo

BEAM NO. 111 DESIGN RESULTS - FLEXURE

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	46.	4- 12 MM	0.	6000.	YES YES
2	543.	4- 20 MM	0.	3343.	YES NO
3	543.	4- 20 MM	2657.	6000.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	441.5/ 1011.4	99.45/ 221.70	227/ 222
500.	414.7/ 761.3	93.41/ 170.24	227/ 222
1000.	379.7/ 538.7	85.53/ 121.34	227/ 222
1500.	333.6/ 337.1	75.15/ 75.93	227/ 222
2000.	289.4/ 234.0	65.18/ 38.71	219/ 230
2500.	234.0/ 234.0	50.66/ 6.85	219/ 230
3000.	234.0/ 0.0	29.76/ 0.00	208/ 0
3500.	234.0/ 234.0	50.43/ 5.44	222/ 227
4000.	292.3/ 234.0	65.83/ 36.99	222/ 227
4500.	334.8/ 322.5	75.42/ 72.64	222/ 227
5000.	380.3/ 518.3	85.65/ 116.73	230/ 219
5500.	416.7/ 735.2	93.85/ 164.75	230/ 219
6000.	444.8/ 979.8	100.19/ 215.33	230/ 219

BEAM NO. 111 DESIGN RESULTS - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	3499 mm	104.6 kN	222	10 mm	10	388 mm
2499	END 2	102.8 kN	219	10 mm	10	388 mm

131J	6000.X 300.X 600	132J
4No20 H 543. 0.TO 3343 4No20 H 543.2657.TO 6000		
10*10c/c388 10*10c/c388		
4No12 H 46. 0.TO 6000		

oooo	oooo	oooo	oooo	oooo	oooo	oooo
4T20	4T20	4T20	4T20	4T20	4T20	4T20
4T12	4T12	4T12	4T12	4T12	4T12	4T12
oooo	oooo	oooo	oooo	oooo	oooo	oooo

BEAM NO. 112 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	48.	4- 16 MM	0.	6500.	YES YES
2	543.	4- 20 MM	0.	3009.	YES NO
3	543.	4- 20 MM	2949.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	442.1/ 1077.7	99.57/ 234.98	227/ 222
542.	425.7/ 774.4	95.88/ 173.01	227/ 222

1083.	399.7/	506.2	90.02/	114.01	227/	222
	378.8/	283.5	85.33/	63.84	219/	230
2167.	347.1/	234.0	78.18/	20.87	219/	230
2708.	284.1/	0.0	63.99/	0.00	219/	0
3250.	289.4/	0.0	65.19/	0.00	222/	0
3792.	405.2/	234.0	91.26/	27.37	222/	227
4333.	486.1/	234.0	109.48/	3.74	222/	227
4875.	535.7/	341.7	120.66/	76.97	222/	227
5417.	562.7/	537.7	126.74/	121.12	230/	219
5958.	603.7/	579.1	135.97/	176.08	230/	219
6501.	635.1/	1072.9	143.04/	234.02	230/	219

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	923.5/	1242.1	203.87/ 267.21
420.	805.0/	984.1	179.40/ 216.20
840.	683.3/	745.0	153.75/ 166.82
1260.	562.8/	533.2	126.76/ 120.10
1680.	446.5/	350.3	100.56/ 78.89
2100.	314.3/	234.0	70.79/ 40.21
2520.	234.0/	234.0	37.02/ 4.78
2940.	234.0/	234.0	33.85/ 7.37
3360.	274.0/	234.0	61.70/ 48.26
3780.	381.8/	407.0	86.00/ 91.66
4200.	487.9/	619.0	109.90/ 139.41
4620.	592.1/	859.4	133.35/ 190.70
5040.	692.1/	1121.3	155.62/ 232.61

1	50.	3- 20 MM	0.	5040.	YES	YES
2	543.	4- 20 MM	0.	5040.	YES	YES

B E A M N O . 1 1 2 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING mm	
mm	kN	kNm	S	T	SIZE	S	T
END 1	4333	116.3	9	222224	10mm	7	21
						29	571
						199	145
4333	END 2	108.9	9	222224	10mm	3	10
						14	571
						199	145

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 402 mm2 EVENLY DISTRIBUTED

* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 402 mm2 EVENLY DISTRIBUTED I

* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *

132J	6500.X 300.X 600	133J
4No20 H 543.1	0.TO 3009	4No20 H 543.2949.TO 6500
29*10c/c145		14*10c/c145
4No16 H 148.	0.TO 6500	
0000	0000	0000
4T20	4T20	4T20
4T16	4T16	4T16
0000	0000	0000

BEAM NO. 113 DESIGN RESULTS - FLEXURE

LEN - 5040. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

136

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B E A M N O. 114 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. mm FC - 40. mm SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	45.	5- 10 MM	0.	6650.	YES YES
2	543.	3- 20 MM	0.	3059.	YES NO
3	543.	3- 20 MM	3591.	6650.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	302.8/	909.1	68.19/ 200.92
554.	326.7/	649.4	73.58/ 146.26
1108.	340.3/	421.0	76.64/ 94.82
1662.	357.5/	234.0	80.52/ 52.23
2217.	366.7/	234.0	82.59/ 17.62
2771.	340.2/	0.0	76.63/ 0.00
3325.	274.8/	0.0	61.90/ 0.00
3879.	288.0/	0.0	64.86/ 0.00
4433.	313.3/	234.0	70.58/ 6.50
4988.	303.0/	234.0	68.25/ 41.32
5542.	285.2/	373.9	64.24/ 84.22
6096.	270.6/	603.4	60.95/ 135.90
6650.	245.7/	859.8	55.35/ 190.79

B E A M N O. 114 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR TORSN mm	LOAD kNm	S	T	LINK SIZE	NO.	SPACING mm	C/C

END 1	3325	100.6	27	224225	10mm	5	16	22	571
3325	END 2	101.0	27	224219	10mm	5	16	22	571

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 547 mm2 EVENLY DISTRIBUTED									
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *									

_____ 126J _____ 6650.X 300.X 600 _____ 122J _____

=====

3No20 H 543. | 0.TO 3059 | | | | 3No20 H 543.3591.TO 6650 | | | |

22*10c/c145 | | | | | | | | 122*10c/c145 | | | |

| | | | | | | | | | | | | | | |

5No10 H |45. | 10.TO 6650 | | | | | | | | | | | |

=====

ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo
3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20
5T10	5T10	5T10	5T10	5T10	5T10	5T10	5T10	5T10	5T10
ooooo	ooooo	ooooo	ooooo	ooooo	ooooo	ooooo	ooooo	ooooo	ooooo

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B E A M N O. 115 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. mm FC - 40. mm SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	4- 20 MM	0.	5310.	YES YES
2	543.	4- 20 MM	0.	2613.	YES NO
3	543.	4- 20 MM	1812.	5310.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	507.0/	1120.5	114.20/ 243.46
442.	440.4/	832.2	99.19/ 185.07
885.	367.8/	570.6	82.83/ 128.51
1328.	286.7/	331.3	64.58/ 74.62
1770.	234.0/	234.0	48.68/ 28.52
2212.	234.0/	0.0	28.99/ 0.00
2655.	287.0/	234.0	64.64/ 6.49
3098.	449.0/	234.0	101.14/ 36.75
3540.	589.7/	313.4	132.81/ 70.59
3982.	714.9/	476.8	160.45/ 107.39
4425.	831.5/	650.2	184.91/ 146.46
4868.	937.4/	843.6	206.71/ 187.43
5310.	1043.6/	1058.9	228.18/ 231.23

=====

B E A M N O. 117 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	65. COMP.	2- 50 MM 4- 10 MM	1317. (REQD. STEEL=	6650. 313. SQ. MM)	NO YES
**Please Note:					
* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*					
2	537. 3- 32 MM	0.	3581.	YES NO	
3	473. 3- 32 MM	0.	3581.	YES NO	
	COMP.	2- 40 MM	(REQD. STEEL=	1627. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)

0.	1221.8/ 4488.4	0.00/ 810.01	0/ 223
554.	380.1/ 3646.7	0.00/ 631.59	0/ 223
1108.	0.0/ 2339.8	0.00/ 457.94	0/ 223
1662.	0.0/ 1370.0	0.00/ 291.63	0/ 223
2217.	247.8/ 602.4	55.80/ 135.68	234/ 223
2771.	645.8/ 234.0	145.45/ 15.64	226/ 231
3325.	1103.8/ 0.0	240.16/ 0.00	225/ 0
3879.	1515.5/ 0.0	318.70/ 0.00	225/ 0
4433.	1874.8/ 0.0	382.34/ 0.00	220/ 0
4988.	2275.9/ 0.0	448.01/ 0.00	220/ 0
5542.	2676.6/ 0.0	507.91/ 0.00	219/ 0
6096.	3332.8/ 234.0	565.06/ 0.00	219/ 0
6650.	3579.9/ 313.3	617.44/ 0.00	219/ 0

B E A M N O. 117 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO mm	SHEAR TORSN kNm	LOAD S	LINK T	NO. S	SPACING mm C/C

END 1 3325	324.2	40	2232223	10mm	13 23 36
3325 END 2	100.4	40	2232221	10mm	5 23 29

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 468 mm2 EVENLY DISTRIBUTED					
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *					

127J	6650.X 300.X 600	123J

3No32 H 537.	10.TO 3581	
36*10c/c 90		29*10c/c111

2No50 H	165.1317.TO 6650	
=====		

OOO	OOO	OOO			
3T32	3T32	3T32			
	2T50	2T50		2T50	2T50
	OO	OO		OO	OO

B E A M N O. 118 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	65. COMP.	2- 50 MM 5- 10 MM	0. (REQD. STEEL=	5310. 347. SQ. MM)	YES YES
2	528. COMP.	2- 50 MM 4- 16 MM	1755. (REQD. STEEL=	5310. 619. SQ. MM)	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)

0.	3613.6/ 347.0	624.57/ 0.00	219/ 0
442.	3345.0/ 234.0	567.65/ 0.00	219/ 0
885.	2738.9/ 0.0	516.71/ 0.00	223/ 0
1328.	2416.0/ 0.0	469.59/ 0.00	223/ 0
1770.	2083.7/ 0.0	417.24/ 0.00	223/ 0
2212.	1733.2/ 0.0	357.81/ 0.00	223/ 0
2655.	1359.5/ 234.0	289.64/ 0.59	223/ 234
3098.	966.0/ 324.5	212.53/ 73.09	223/ 234
3540.	622.5/ 737.4	140.20/ 165.21	232/ 225
3982.	383.2/ 1349.9	86.32/ 287.82	232/ 225
4425.	234.0/ 2074.0	28.75/ 415.66	232/ 225
4868.	0.0/ 2966.4	0.00/ 547.71	0/ 225
5310.	619.4/ 3886.0	0.00/ 682.31	0/ 225

B E A M N O. 120 D E S I G N R E S U L T S - F L E X U R E									
LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm									
LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END				

1	65.	2- 50 MM	1317.	6650.	NO	YES			
	COMP.	5- 10 MM	(REQD. STEEL=	340. SQ. MM)					
*Please Note:									
Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*									
2	537.	3- 32 MM	0.	3581.	YES	NO			
3	473.	3- 32 MM	0.	3581.	YES	NO			
	COMP.	2- 40 MM	(REQD. STEEL=	1676. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

S E C T I O N R E I N F S T E E L (+ V E / - V E) M O M E N T S (+ V E / - V E) L O A D (+ V E / - V E)									
(M M) (S Q. M M) (K N - M E T E R)									
0.	1267.4/	4534.0	0.00/	819.69	0/	224			
554.	418.5/	3685.1	0.00/	639.74	0/	224			
1108.	0.0/	2382.9	0.00/	464.55	0/	224			
1662.	0.0/	1397.0	0.00/	296.71	0/	224			
2217.	258.5/	618.2	58.22/	139.23	233/	224			
2771.	659.1/	234.0	148.45/	18.07	225/	232			
3325.	1121.0/	0.0	243.55/	0.00	226/	0			
3879.	1536.5/	0.0	322.55/	0.00	226/	0			
4433.	1891.4/	0.0	385.17/	0.00	226/	0			
4988.	2297.1/	0.0	451.32/	0.00	222/	0			
5542.	2704.7/	0.0	511.90/	0.00	221/	0			
6096.	3355.7/	234.0	569.91/	0.00	221/	0			
6650.	3606.9/	340.3	623.16/	0.00	221/	0			

B E A M N O. 120 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR	TORSN	LOAD	LINK	NO.				SPACING mm				C/C					
	mm	kN	kNm	S	T	SIZE	S	T	S+T	S	T	S+T	S	T	S+T				

END 1	3325	327.0	38	224224	10mm	13	22	36	249	147	92								
3325 END 2	103.2	38	224219	10mm	5	22	28	571	147	115									

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 467 mm2 EVENLY DISTRIBUTED																			
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *																			

128J										6650.X 300.X 600										124J									
=====																													
3No32 H 537.										10.TO 3581																			
36*10c/c 92																				28*10c/c115									
2No50 H										165.1317.TO 6650																			
=====																													
OOO										OOO										OOO									
3T32										3T32										3T32									
										2T50										2T50									
										OO										OO									

B E A M N O. 121 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING mm	C/C
mm	kN	S T	SIZE	S T	S T	S T
END 1 2655	161.8	45	226226	10mm	4	21 26 571 126 102
2654 END 2	311.2	45	226219	10mm	10	21 31 256 126 83
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 633 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

124J	5310.X 300.X 600	120J
26*10c/c102		
2No50 H 65. 0.TO 5310		
2No40 H 533.1844.TO 5310		
31*10c/c 83		

2T50	2T50	2T50	2T50	2T50	2T50	2T50
OO	OO	OO	OO	OO	OO	OO

B E A M N O. 122 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	50.	3- 20 MM	0.	5040.	YES	YES
2	541.	3- 25 MM	0.	5040.	YES	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	918.8/ 1259.8	202.91/ 270.62	234/ 223
420.	802.5/ 1003.1	178.88/ 220.04	234/ 223
840.	683.0/ 765.3	153.68/ 171.08	234/ 223

1260.	563.0/	552.6	126.81/	124.45	234/ 223
1680.	440.7/	363.6	99.26/	81.89	226/ 231
2100.	310.6/	234.0	69.96/	43.64	226/ 231
2520.	234.0/	234.0	36.65/	8.66	225/ 232
2940.	234.0/	234.0	27.45/	5.21	223/ 234
3360.	243.6/	234.0	54.88/	45.65	223/ 234
3780.	353.4/	397.2	79.60/	89.46	231/ 226
4200.	463.0/	612.5	104.28/	137.97	231/ 226
4620.	565.2/	850.2	127.29/	188.80	231/ 226
5040.	662.1/	1109.4	149.12/	241.26	231/ 226

B E A M N O. 122 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1 2519 mm	121.8 kN	223	10 mm	8	359 mm
2519 END 2	126.3 kN	226	10 mm	8	359 mm

133J	5040.X 300.X 600	129J
3No25 H 541. 0.TO 5040		
8*10c/c359		
3No20 H 50. 0.TO 5040		

OOO	OOO	OOO	OOO	OOO	OOO
3T25	3T25	3T25	3T25	3T25	3T25
3T20	3T20	3T20	3T20	3T20	3T20
OOO	OOO	OOO	OOO	OOO	OOO

B E A M N O. 123 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	45.	5- 10 MM	0.	6650.	YES	YES
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2	543.	3- 20 MM	0.	3059.	YES	NO
3	543.	3- 20 MM	3591.	6650.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	295.4/ 894.7	66.53/ 197.97	234/ 223
554.	317.6/ 634.1	71.54/ 142.82	234/ 223
1108.	329.5/ 403.4	74.22/ 90.87	234/ 223
1662.	349.2/ 234.0	78.65/ 48.72	226/ 231
2217.	356.7/ 234.0	80.35/ 13.61	226/ 231
2771.	328.6/ 0.0	74.01/ 0.00	226/ 0
3325.	261.5/ 0.0	58.91/ 0.00	226/ 0
3879.	315.9/ 0.0	71.15/ 0.00	224/ 0
4433.	343.4/ 234.0	77.34/ 11.03	223/ 234
4988.	335.3/ 234.0	75.53/ 46.24	223/ 234
5542.	315.5/ 393.2	71.06/ 88.56	231/ 226
6096.	303.1/ 624.3	68.28/ 140.63	231/ 226
6650.	280.5/ 884.6	63.17/ 195.90	231/ 226

B E A M N O. 123 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING	mm C/C
mm	KN	S T	SIZE	S T	S T	S T
END 1 3325	101.5	31	223226	10mm	5 18 25	571 180 133
3325 END 2	101.7	31	223221	10mm	5 18 25	571 180 133
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 597 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

129J	6650.X 300.X 600	125J
3No20 H 543.	10.TO 3059	3No20 H 543.3591.TO 6650
25*10c/c133		125*10c/c133
5No10 H 145.	10.TO 6650	

ooo	ooo	ooo	ooo	ooo	ooo
3T20	3T20	3T20	3T20	3T20	3T20
5T10	5T10	5T10	5T10	5T10	5T10
ooooo	ooooo	ooooo	ooooo	ooooo	ooooo

B E A M N O. 124 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	50.	4- 20 MM	0.	5310.	YES YES
2	543.	4- 20 MM	0.	2613.	YES NO
3	543.	4- 20 MM	1812.	5310.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	549.1/ 1178.4	123.67/ 254.84	234/ 223
442.	480.1/ 884.0	108.13/ 195.76	234/ 223
885.	405.1/ 615.0	91.24/ 138.53	234/ 223
1328.	321.7/ 372.8	72.46/ 83.96	234/ 223
1770.	244.5/ 234.0	55.08/ 36.23	226/ 231
2212.	234.0/ 0.0	34.88/ 0.00	226/ 0
2655.	254.6/ 234.0	57.35/ 0.20	224/ 233
3098.	419.7/ 234.0	94.54/ 31.00	224/ 233
3540.	563.3/ 290.2	126.87/ 65.36	223/ 234
3982.	690.1/ 455.9	155.19/ 102.69	223/ 234
4425.	809.5/ 631.7	180.33/ 142.29	223/ 234
4868.	918.3/ 826.1	202.81/ 183.79	223/ 234
5310.	1031.3/ 1047.1	225.71/ 228.88	231/ 226

B E A M N O. 124 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING	mm C/C
mm	KN	kNm	S T	SIZE	S T	S T
END 1 2655	135.0	41	223231	10mm	4 19 23	571 138 111
2654 END 2	107.4	41	223221	10mm	4 19 23	571 138 111
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 762 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

